RT58X MPTCB and RT584 MP Tool

User Guide

V2.1

Introduction

This document describes the RT584 Mass Production Pack developed by Rafael Microelectronics. The goal of the Mass Production Pack is to provide a utility for testing an “RT584-inside” product’s functionality on production lines and without the need for additional equipment.

The Mass Production Pack includes the following hardware and software items:

- **1-to-4 Mass Production Test Control Board (MPTCB)** – hardware providing Bluetooth®, Zigbee 3.0 & Sub-GHz TX/RX capabilities controlled by the MP Tool program
- **Mass Production Tool (MP Tool)** – software application running on the Microsoft Windows operating system

The following features are included in this pack:

- Over Current Protection (OCP) Check
- Over Voltage Protection (OVP) Check
- Erase/ Write/ Verify DUT Flash
- GPIO test
- Crystal Trim
- RSSI Test
- RF Tx Test
- RF Rx Test
- Secure Bootloader Setting
- Write Matter Certificate

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1. Environment Setup for Mass Production Test Control Board (MPTCB)

The complete testing setup includes:

- One PC (running Windows XP operating system or later version)
- 1-to-4 MPTCB hardware (from Rafael Micro)
- One Type-B USB cable
- One 5V/3A Adapter (from Rafael Micro)
- MP Tool application software (from Rafael Micro)

Figure 1 and **Figure 2** show a 1-to-4 MPTCB hardware. The user must connect the DC 5V/3A adapter to power the MPTCB. To avoid any instability caused by power supply issues, we suggest using the 5V/3A adapter from Rafael Micro.

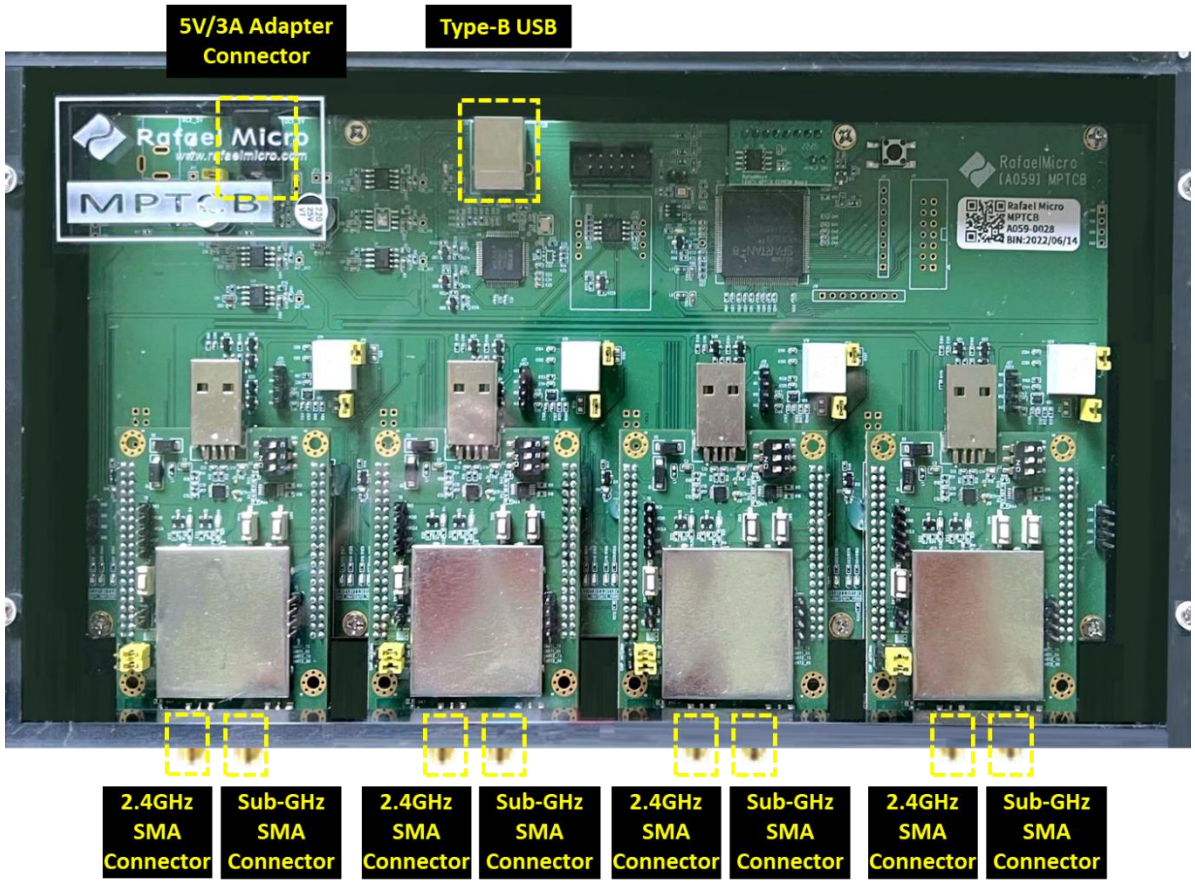


Figure 1. Connectors on 1-to-4 MPTCB (Top View)

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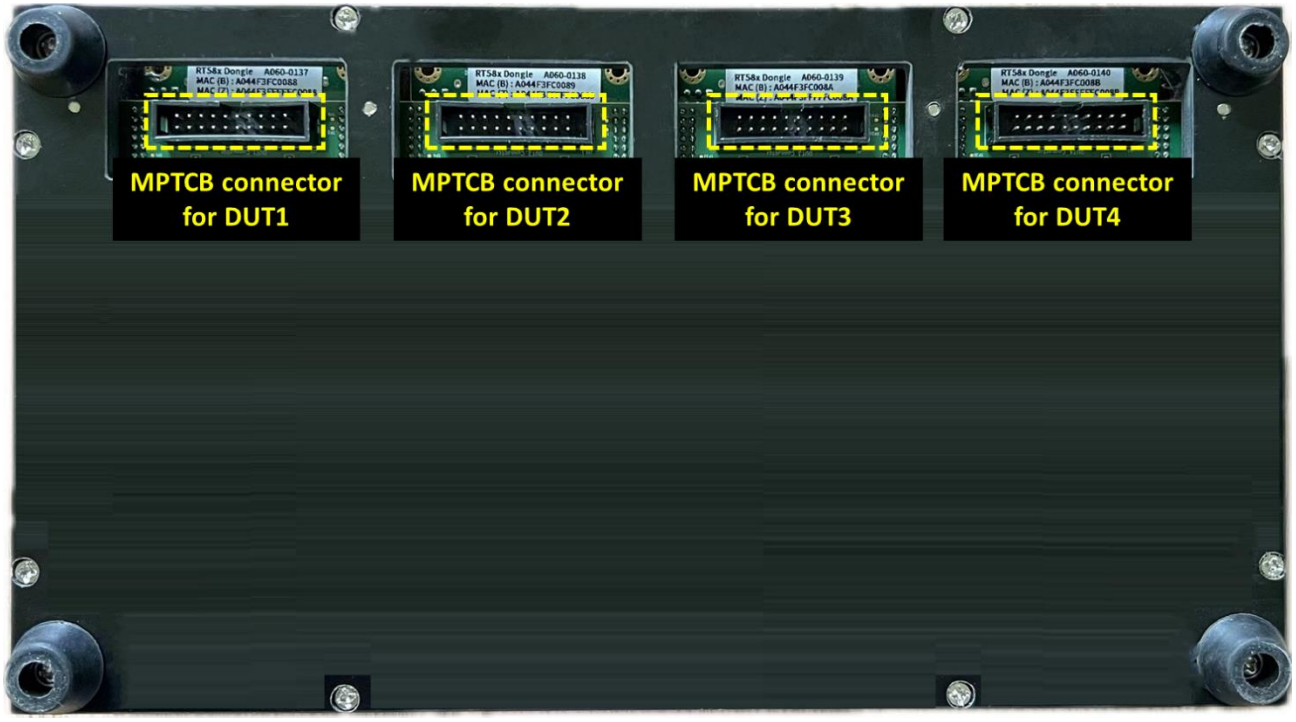


Figure 2. Connectors on 1-to-4 MPTCB (Bottom View)

Figure 3 illustrates a reference connection setup between the PC, the 1-to-4 MPTCB, and the DUT (Device Under Test). Ensure all connections are secure before powering on the system.

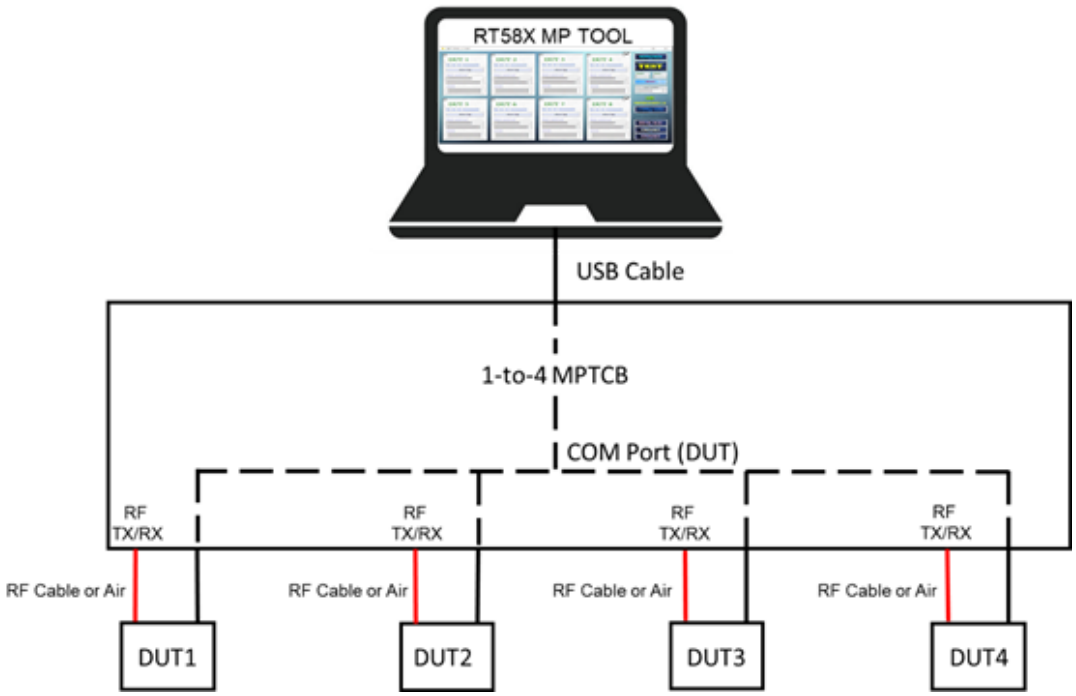


Figure 3. 1 to-4 MPTCB Connection block diagram

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1.1 Hardware setup of Mass Production Test Control Board (MPTCB)

The 1-to-4 MPTCB features one Type-B USB port for connection to the PC. **Before connecting the USB cable to the PC, the DC 5V/3A adapter must be connected to power the MPTCB.** A USB-to-UART bridge IC is integrated with the Type-B USB port on the MPTCB.

When the USB connection is established, four virtual UART ports will appear as USB-to-Serial bridges in the Windows Device Manager. These COM ports are used in the software settings to identify and communicate with individual devices.

The **MPTCB connector,** as shown in **Figure 4**, serves as a UART interface that links the DUT (Device Under Test) to the PC. In addition to data transmission, it supplies VCC to the DUT, eliminating the need for an external power source. The DUT's reference clock is also transmitted to the MPTCB via this interface for crystal trimming purposes.

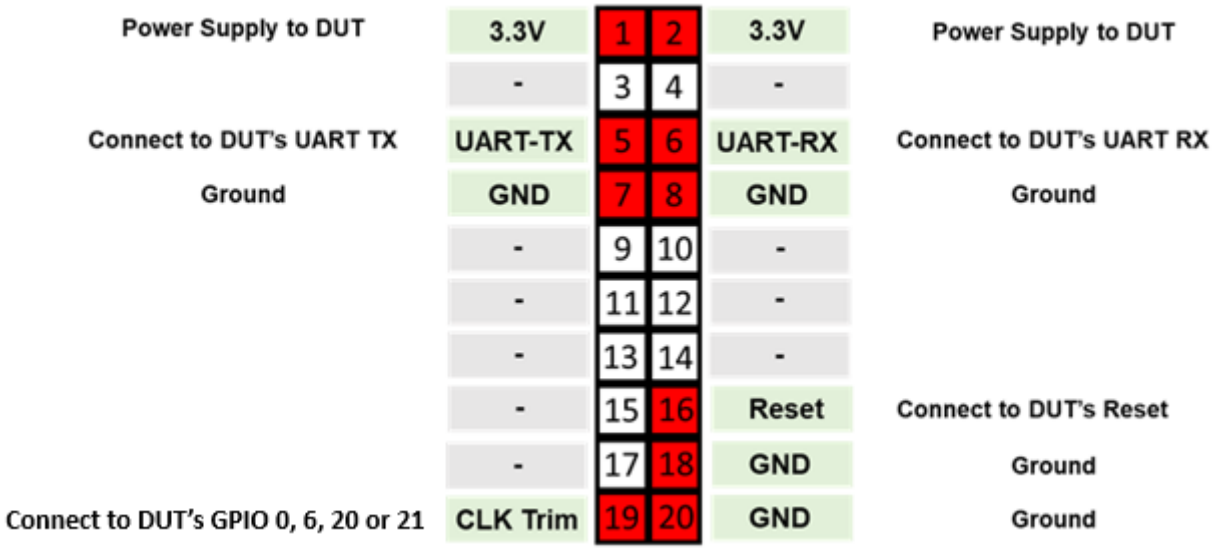


Figure 4. MPTCB Connector Hardware Description

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1.2 Power supply of 1-to-4 Mass Production Test Control Board (MPTCB)

Power is supplied to the 1-to-4 MPTCB using a 5V/3A DC power adapter. **It is important to connect the DC 5V/3A adaptor to the MPTCB before connecting the Type-B USB port to the PC.**

To avoid instability caused by power-related issues, **it is strongly recommended to use the 5V/3A adapter provided by Rafael Micro** when operating the 1-to-4 MPTCB.

- **DC 5V/3A** – Use a DC 5V/3A adapter to supply power to the 1-to-4 MPTCB.
- **VCCIO** – The MP Tool can automatically select either 1.8V or 3.3V for VCCIO on the 1-to-4 MPTCB. However, **VCCIO must be manually set** to 1.8V or 3.3V when using the RT584 Evaluation Kit (RT584 EVK).

Figure 5 shows the power connection between the 1-to-4 MPTCB and RT584 EVK.



Figure 5. Connect RT584 EVK to 1-to-4 MPTCB

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Figure 6 and **Table 1** show jumper settings for the RT584 EVK when connected to the 1-to-4 MPTCB. The EVK includes a connector for interfacing with the **Rafael Micro MPTCB** using a 20-pin IDE cable. The Rafael Micro MPTCB serves as a calibration kit used for crystal trimming and firmware downloading to Rafael Micro chipsets. When the RT584 EVK is connected to the MPTCB, **both SW8 and SW9 must be set to OFF** to ensure proper operation.

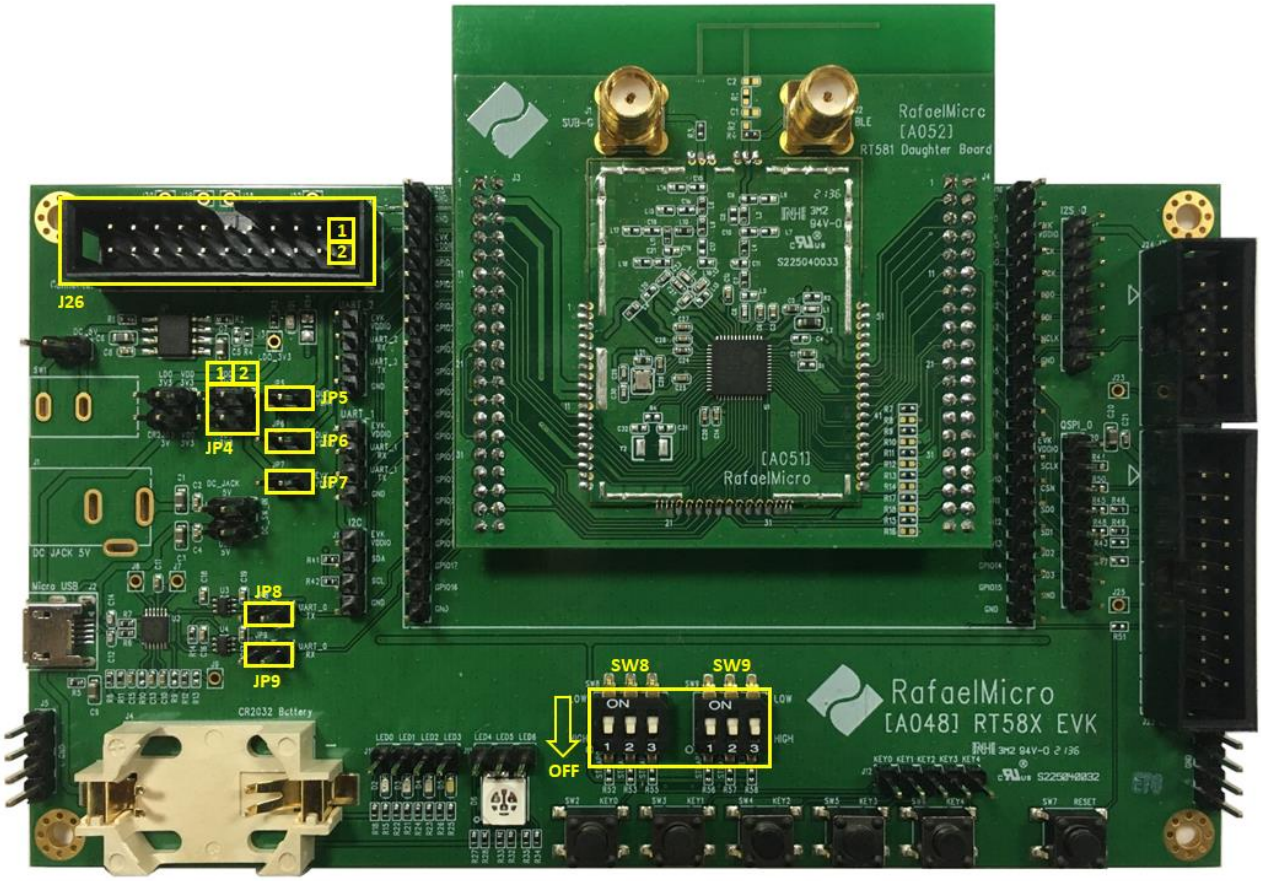


Figure 6. Top view of the RT58X Series Evaluation Board

Table 1. MPTCB connection jumper settings

Mode	JP4	JP5	JP6	JP7	JP8	JP9	SW8	SW9
MPTCB connection	X	O	O	X	X	X	OFF	OFF

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1.3 Software setup of the Mass Production Test Control Board (MPTCB)

The RT584 1-to-4 Mass Production Test Control Board (MPTCB) includes a built-in USB-to-serial interface. When the MPTCB is connected to the PC using a Type-B USB cable, **four COM ports will be automatically detected** by the Windows operating system.

Among the four COM ports, **UART0 is designated for console command input**.

Figure 7 shows a reference connection diagram for the software setup.

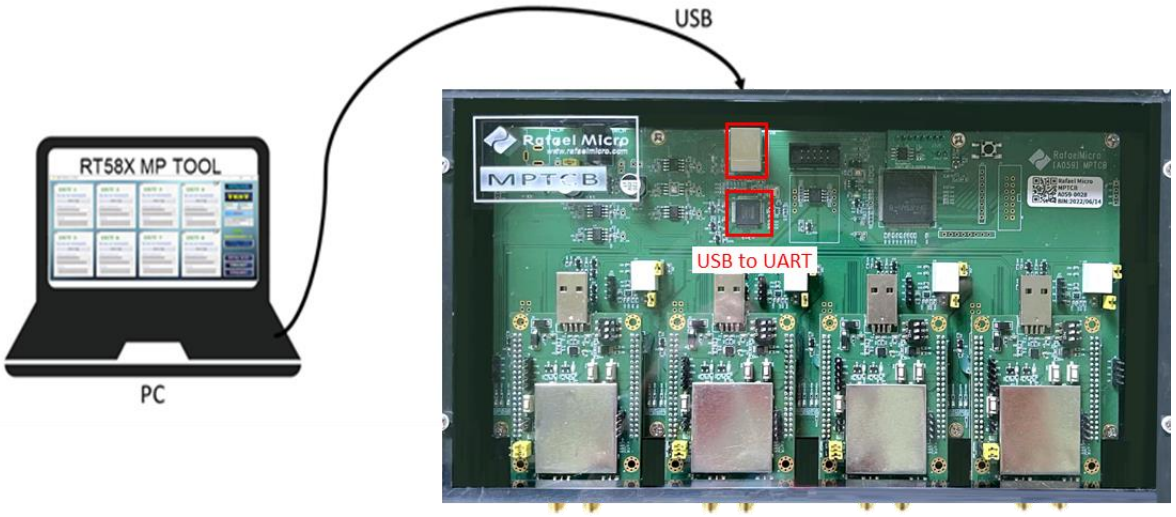


Figure 7. RT58X MPTCB Connections

FT4232HL Driver Installation for Windows

Installation of the FT4232HL driver is requires for windows. The FTDI driver for the FT4232HL chip used on the on the RT58X MPTCB can be downloaded from the following link:

https://ftdichip.com/wp-content/uploads/2021/08/CDM212364_Setup.zip

Once the driver is installed and the MPTCB is connected to the PC, **four COM ports will appear** in the Windows Device Manager (as shown in **Figure 8**).

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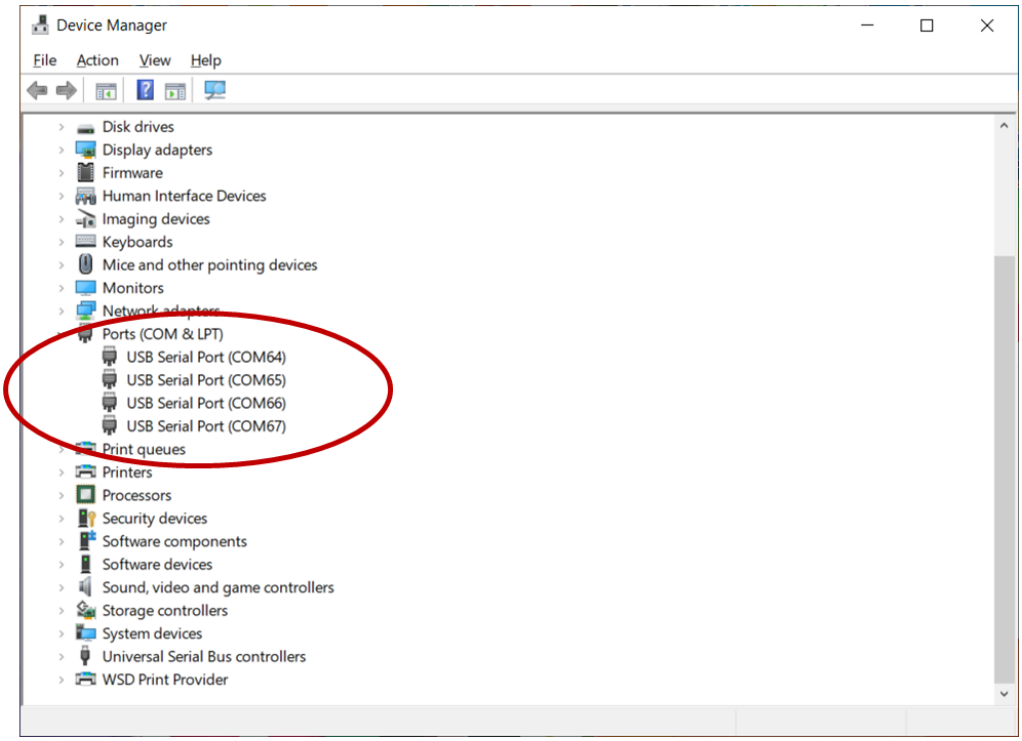


Figure 8. Windows Device Manager showing four COM ports with the FTDI driver installed and the 1-to-4 MPTCB powered and connected.

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2. Mass Production Tool User Interface

The **Mass Production Tool (MP Tool)** is used for performing mass production tests on devices incorporating the Rafael Micro RT584 chip. The mass production process includes **flash write/verify**, **crystal trimming**, and **RF TX/RX testing**. Once the installation is complete, you can launch the application by double-clicking **"MP_TOOL.exe"** in the installation directory. The user interface upon startup is shown in **Figure 9**.

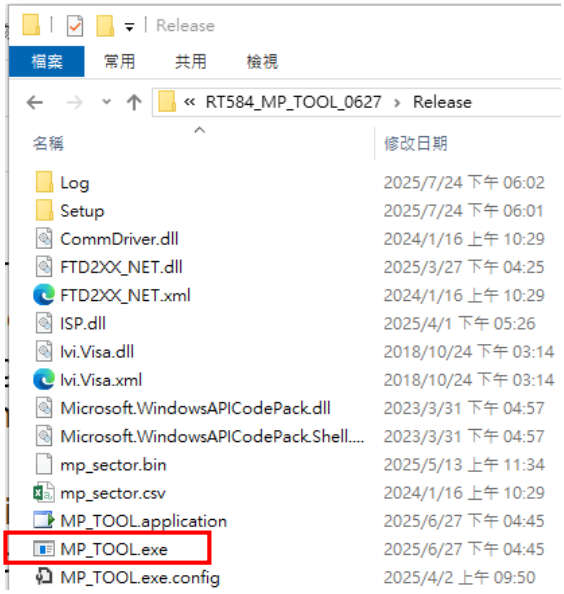


Figure 9. RT584 MP Tool Software Path

The **MP Tool** generates a separate log file for each DUT. These log files are saved in the designated working folder, as shown in **Figure 10**.

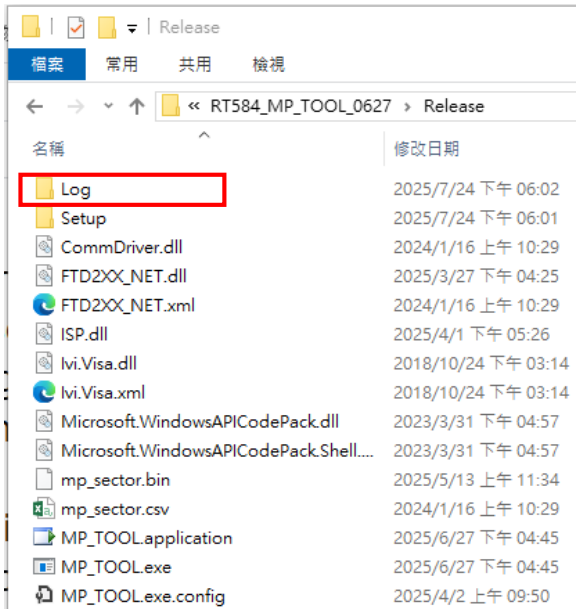


Figure 10. Log list in file manager

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Figure 11 shows the graphical user interface (GUI) of the MP Tool.

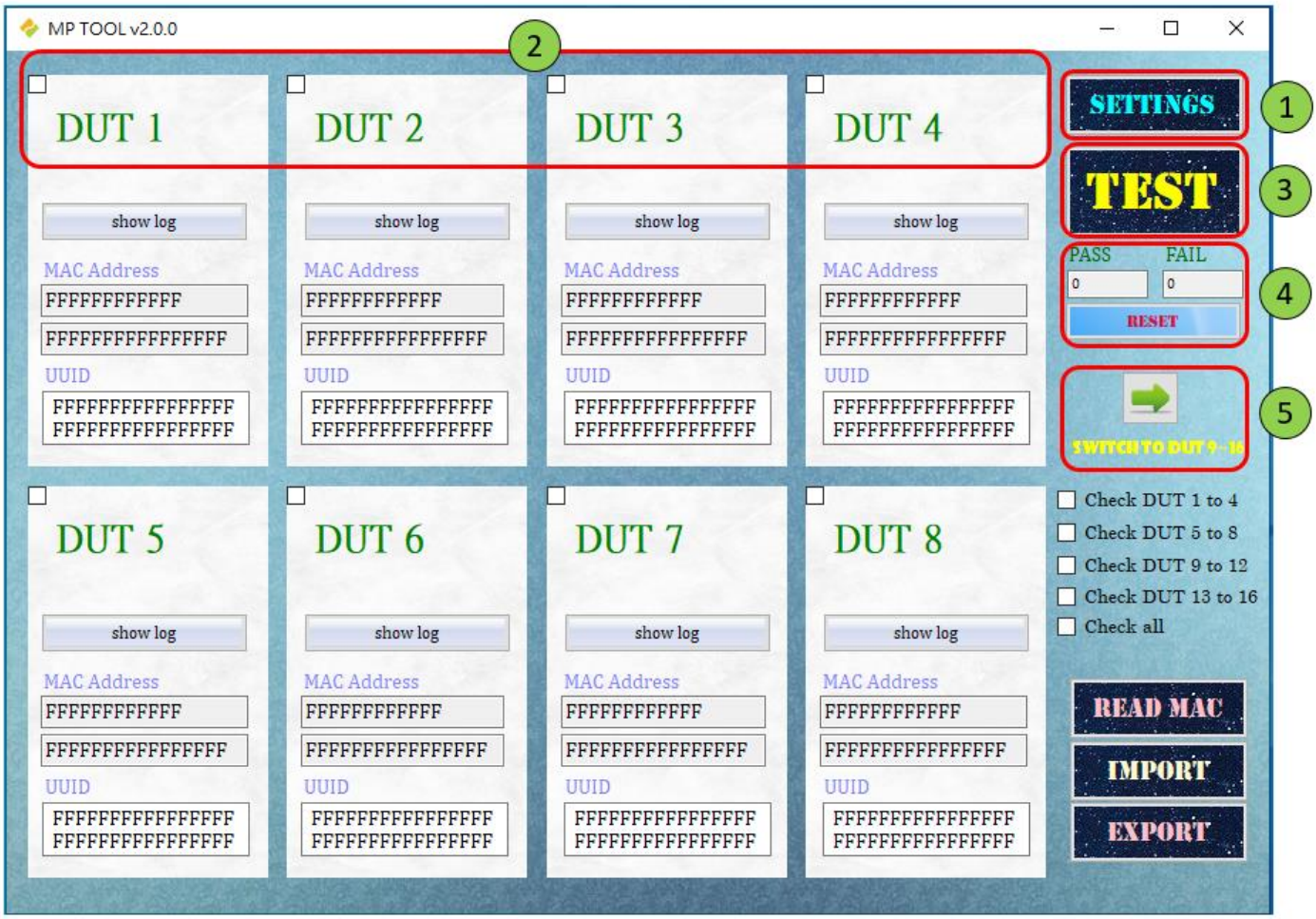


Figure 11. Tool setup progress displayed during initialization.

Quick Start Procedure:

1. Set the test parameters in the **Setting** panel for the selected test item(s).
2. Select the DUT(s) for testing by checking the corresponding checkbox. The DUTs are listed in the same order as the COM ports. Click the **TEST** button to begin testing.
3. After testing, MP Tool will automatically update the **Pass/Fail** counts.
4. MP Tool supports **1 to 16 parallel tests**. Click the toggle button to switch between the two display windows.

If a DUT passes all tests, its corresponding box will show **PASS** in **green**, as illustrated in **Figure 12**.

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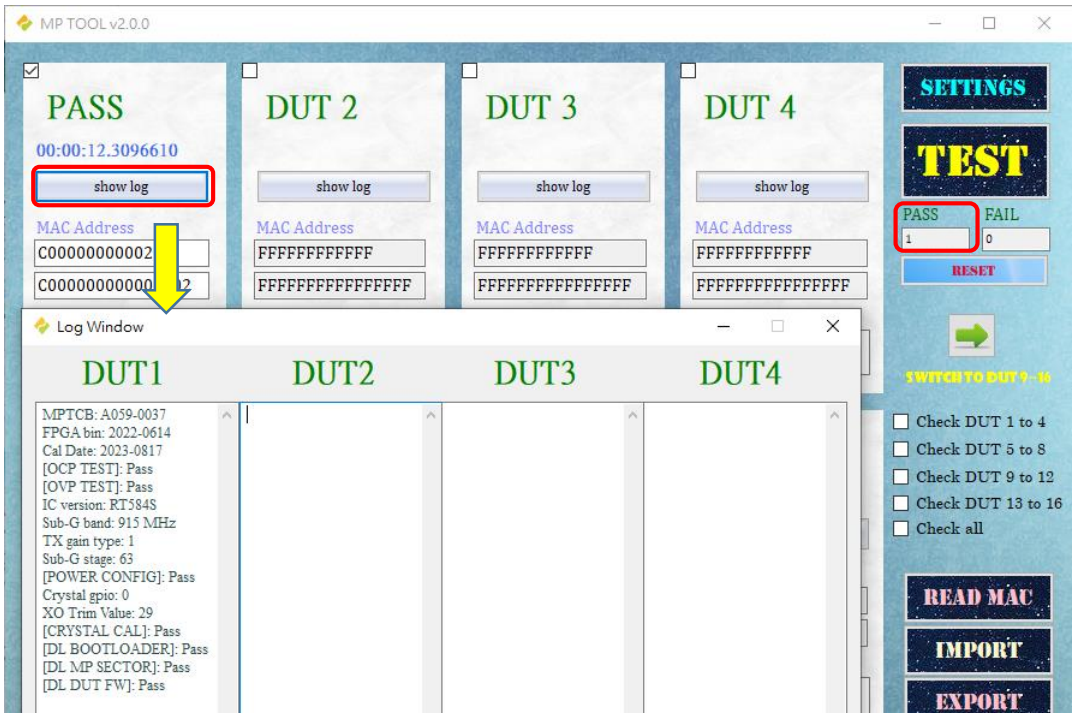


Figure 12. Test pass result

If a DUT fails a test, the corresponding DUT box will display **FAIL** in red**,** as shown in **Figure 13**.

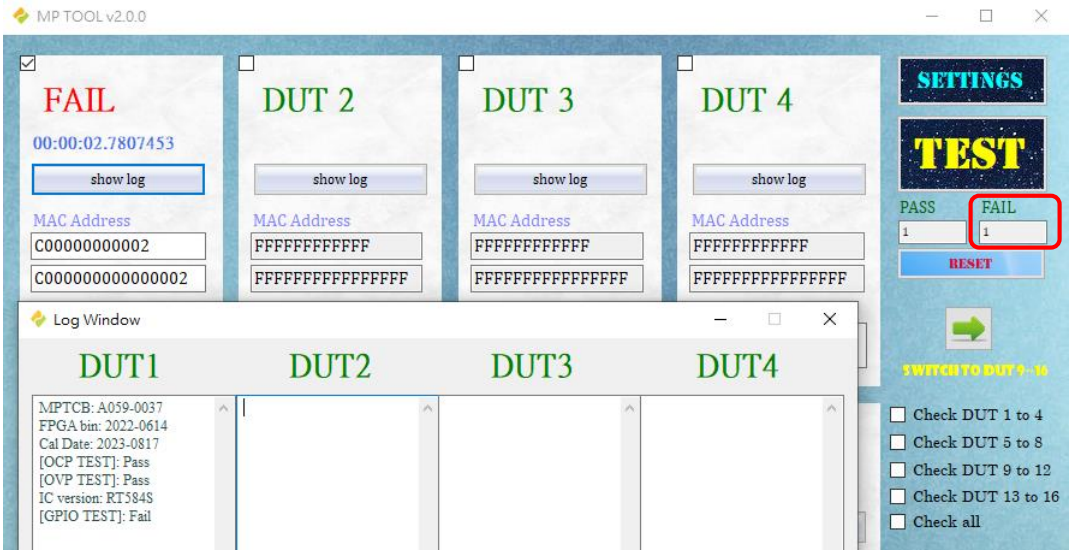


Figure 13. Test fail result

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3. Test Configuration

3.1 Download Dongle/ Test/ DUT/ Bootloader Firmware to the MPTCB.

The operation procedure of the MP Tool is described below. (See **Figure 14**)

1. Click the **Sync** button to confirm that the DUT COM port settings are correct.
2. Select the appropriate **test firmware** based on the test chip.
3. Click the **Select DUT Image** to select the DUT image.
   - If no DUT image is selected, MP Tool will display a warning message such as: “DUT FW does not exist!”
4. Click the **Select Bootloader Image** to select the bootloader image.
   - If no bootloader image is selected, MP Tool will display a warning message such as: “Bootloader FW does not exist!”
5. Click the **Download** to download the dongle firmware from PC to the MPTCB.
6. Click the **Download** to download the Test/ DUT/ Bootloader firmware from PC to the MPTCB.
7. Click the **Test Config** button to configure the detailed settings for each test item.
8. Select the checkboxes for the test items.
   - Note: User are not allowed to deselect any item in the to-do list.

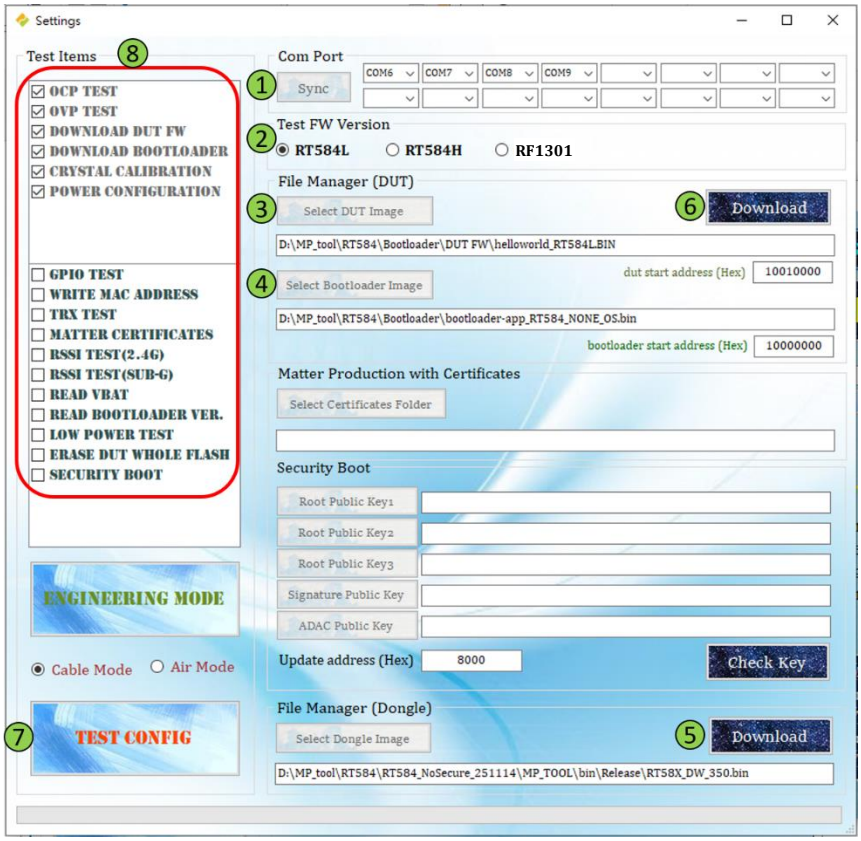


Figure 14. Setting dialog

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3.2 Over Current Protection (OCP) Test

To verify whether the Device Under Test (DUT) has a short circuit that results in excessive current, the MPTOOL provides an Over Current Protection (OCP) test function. This function allows users to set a current limit to screen out potentially defective DUTs.

As shown in **Figure 15**, users can select the desired current limit from the dropdown menu in the lower-left corner (e.g., 50 mA, 100 mA, 150 mA, etc.). If the DUT's current exceeds the configured threshold, the system will automatically mark the DUT test failed.

The OCP test settings in the MP Tool are described below: (See Figure 15~18)

1. Go to the ConfigForm page and select the desired current limit.
2. Go to the settings page and enable **OCP TEST**.
3. Go to the MP TOOL v2.0.0 page and press the **TEST**.
4. Press **show log** to view the result.

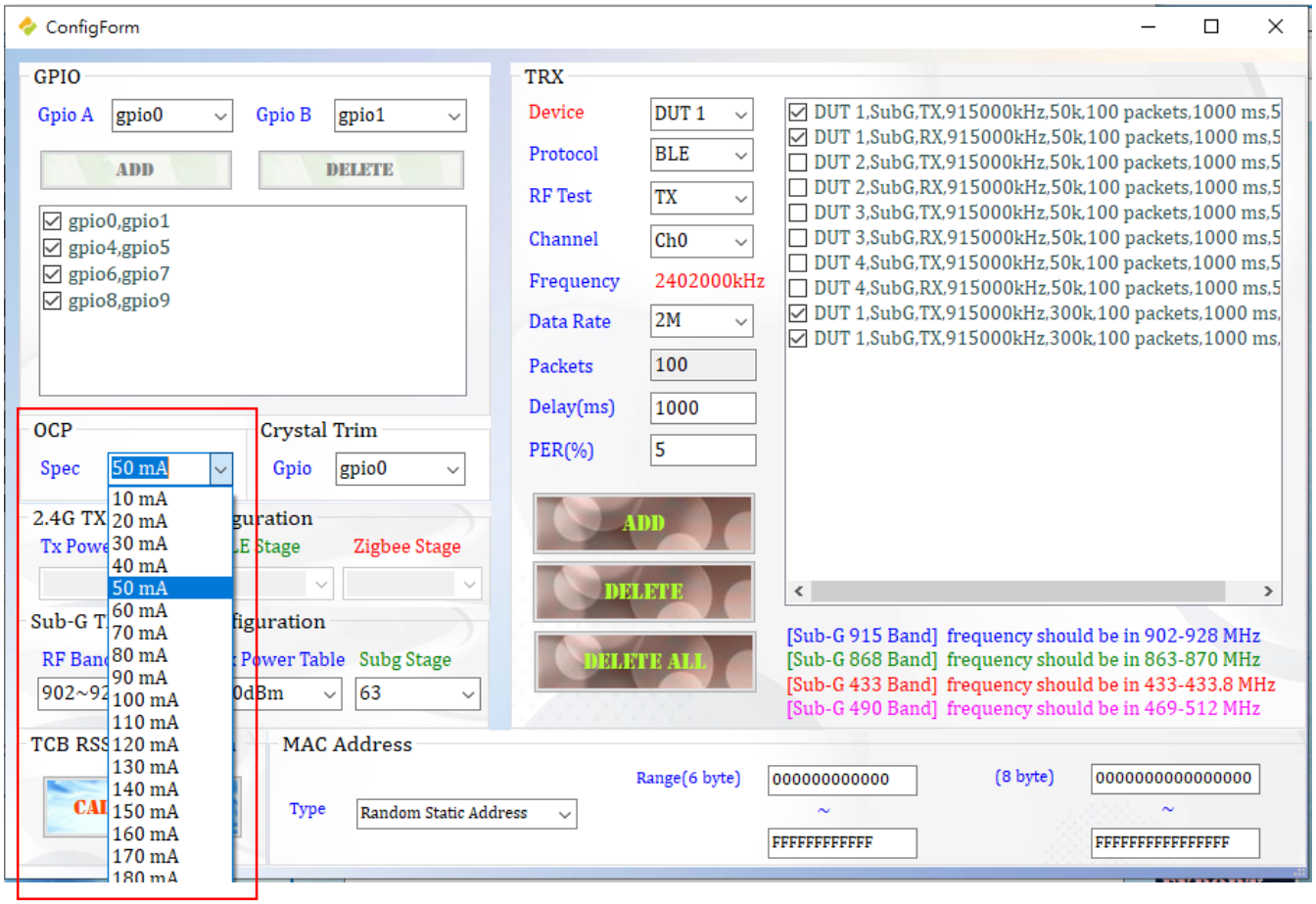


Figure 15. Over Current Protection (OCP) current limit Setting

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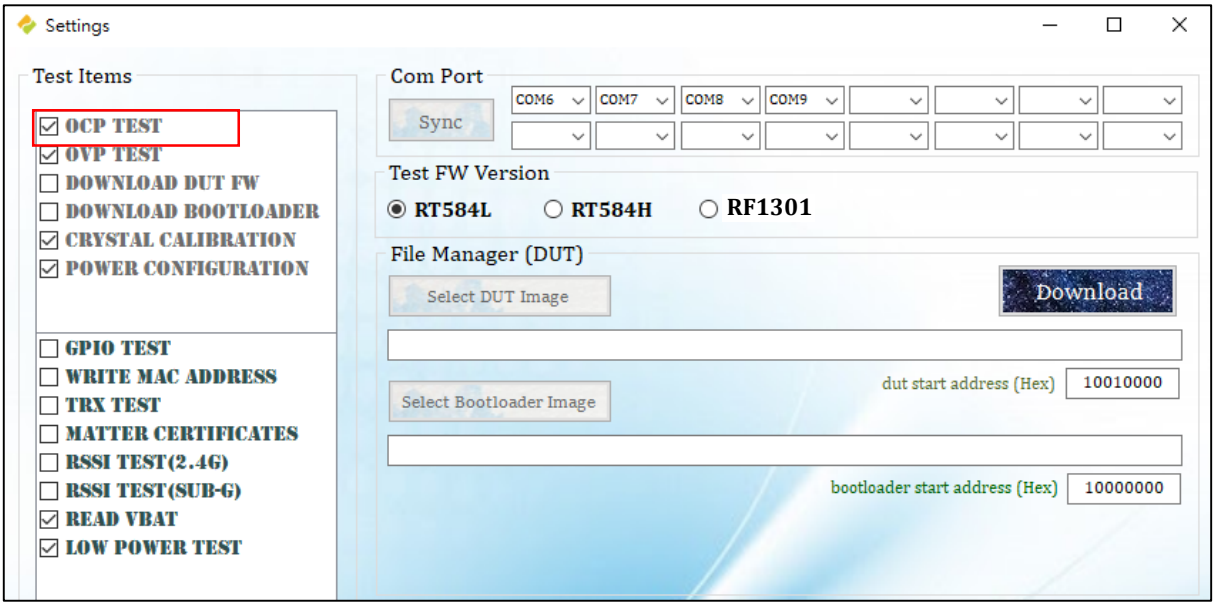


Figure 16. Check the test item - OCP TEST.

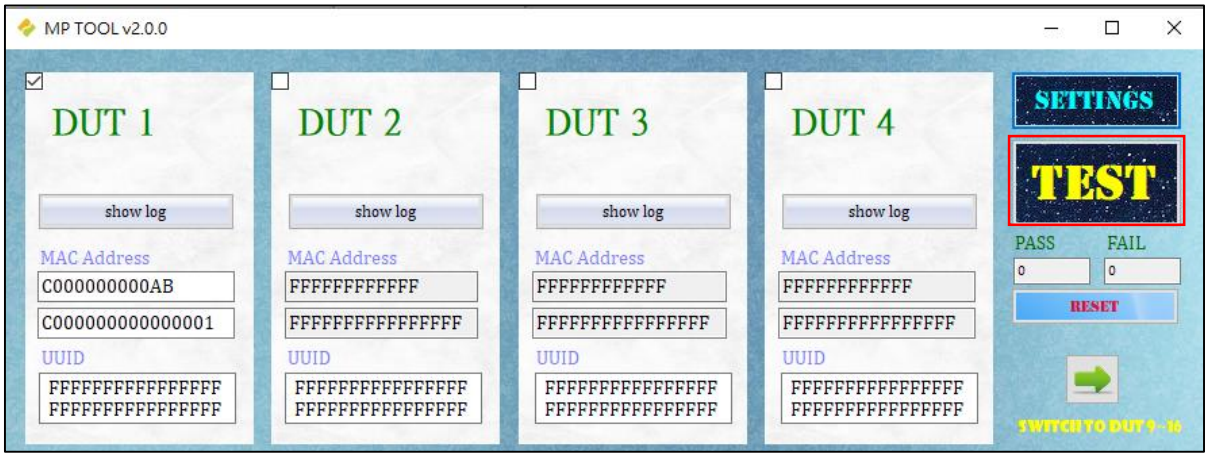


Figure 17. Click the TEST button.

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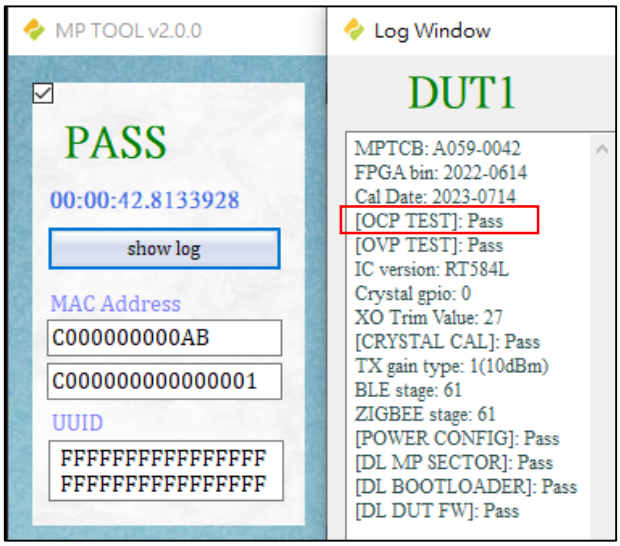


Figure 18. Show the log of the OCP TEST.

3.3 Over Voltage Protection (OVP) Test

This test verifies whether the DUT supports the over voltage protection (OVP) function properly.

The OVP test settings in the MP Tool are described below: (See Figure 19~21)

1. Go to the settings page and enable **OVP TEST**.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

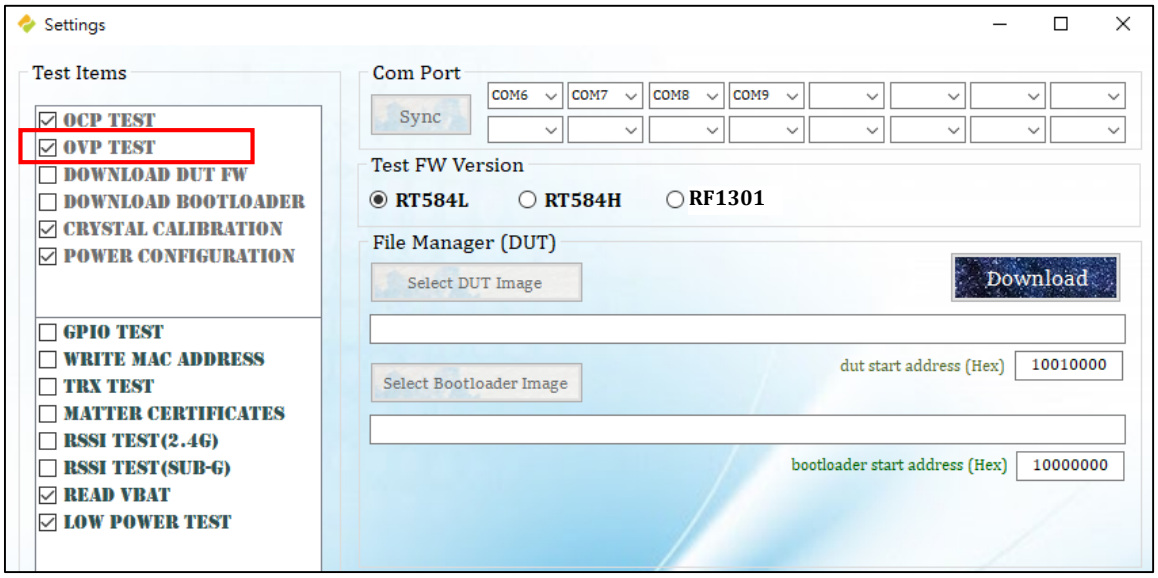


Figure 19. Check the test item - OVP TEST.

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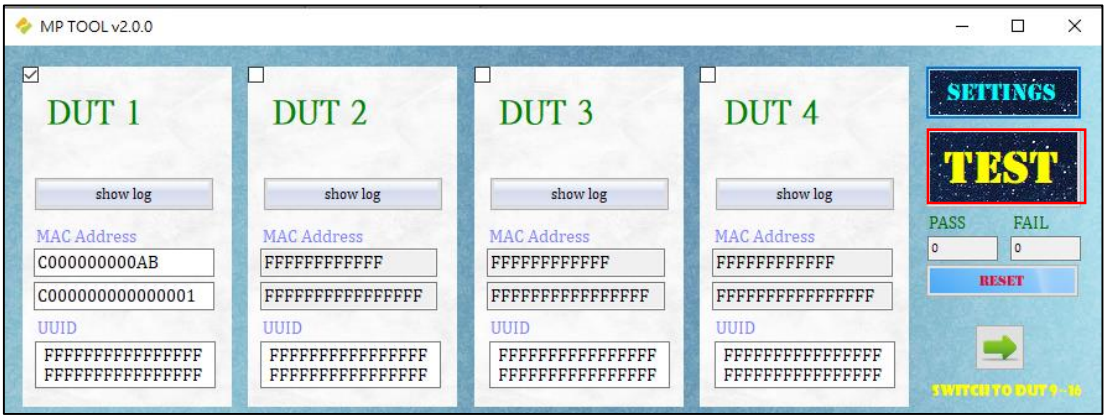


Figure 20. Click the TEST button.

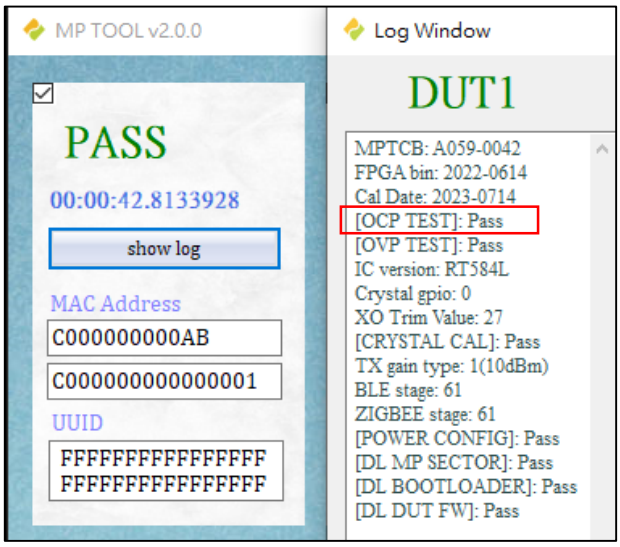


Figure 21. Show the log of the OCP TEST.

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3.4 Download DUT Firmware to the DUT

The procedure for downloading DUT firmware to the DUT in the MP Tool is described below: (See Figure 22~23).

1. **Perform the steps in 3.1 Download Dongle / TEST/ DUT/ Bootloader Image to the MPTCB**.
   - If MPTCB does not contain the dongle, test and DUT firmware, the MP tool will fail to download the DUT firmware to the DUT.
2. Go to the settings page and enable **DOWNLOAD DUT FW**.
3. Go to the MP TOOL v2.0.0 page and press the **TEST**.
4. Press **show log** to view the result.

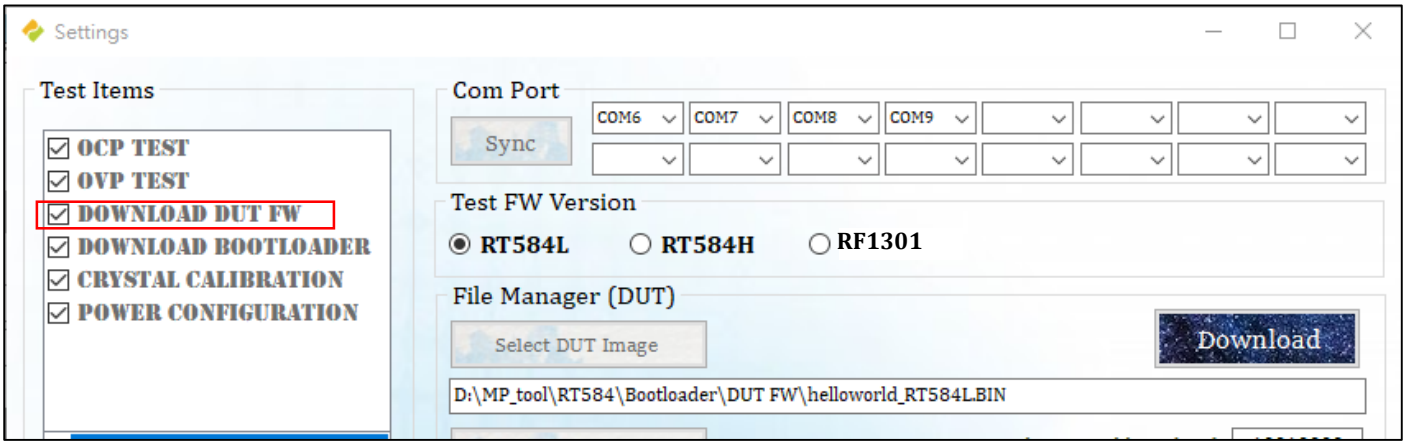


Figure 22. Check the test item - DOWNLOAD DUT FW.

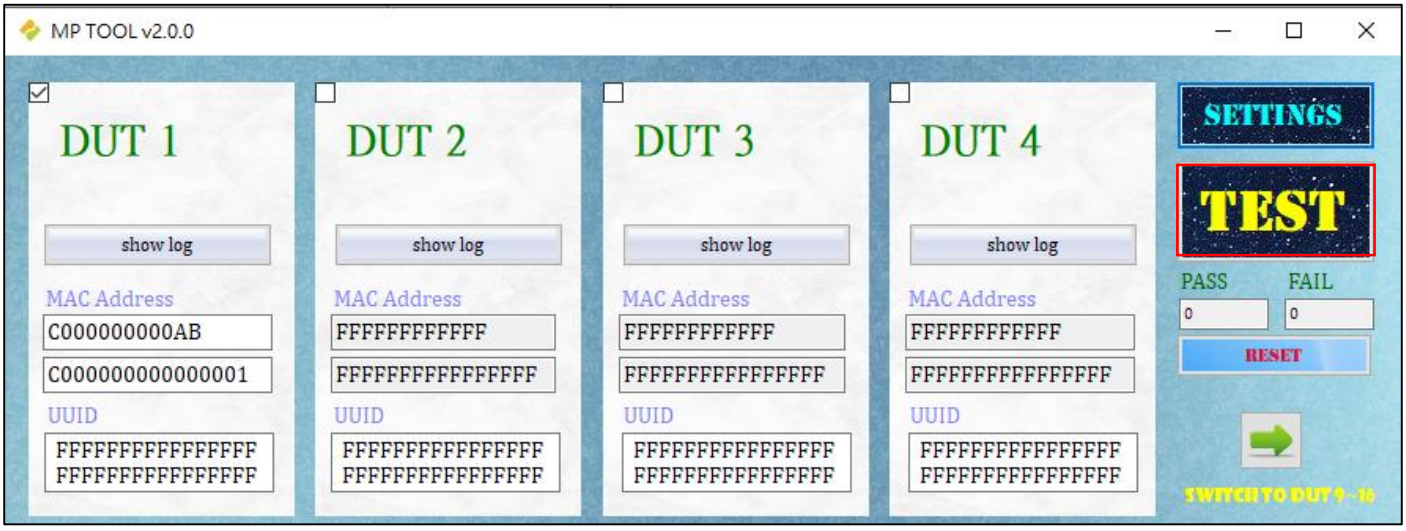


Figure 23. Click the TEST button.

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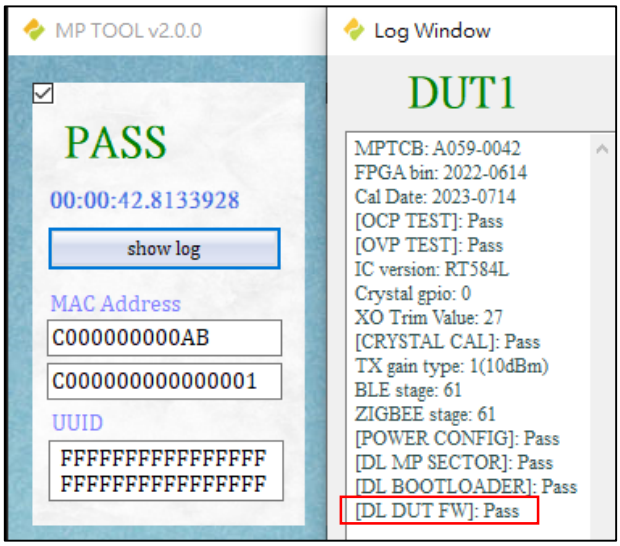


Figure 24. Show the log of the DOWNLOAD DUT FW.

3.5 Download Bootloader Firmware to the DUT

The procedure for downloading bootloader firmware to the DUT in the MP Tool is described below: (See Figure 25~27).

1. **Perform the steps in 3.1 Download Dongle / TEST/ DUT/ Bootloader Firmware to the MPTCB**.
   - If MPTCB does not contain the dongle, test and bootloader firmware, the MP tool will fail to download the DUT firmware to the DUT.
2. Go to the settings page and enable **DOWNLOAD BOOTLOADER**.
3. Go to the MP TOOL v2.0.0 page and press the **TEST**.
4. Press **show log** to view the result.

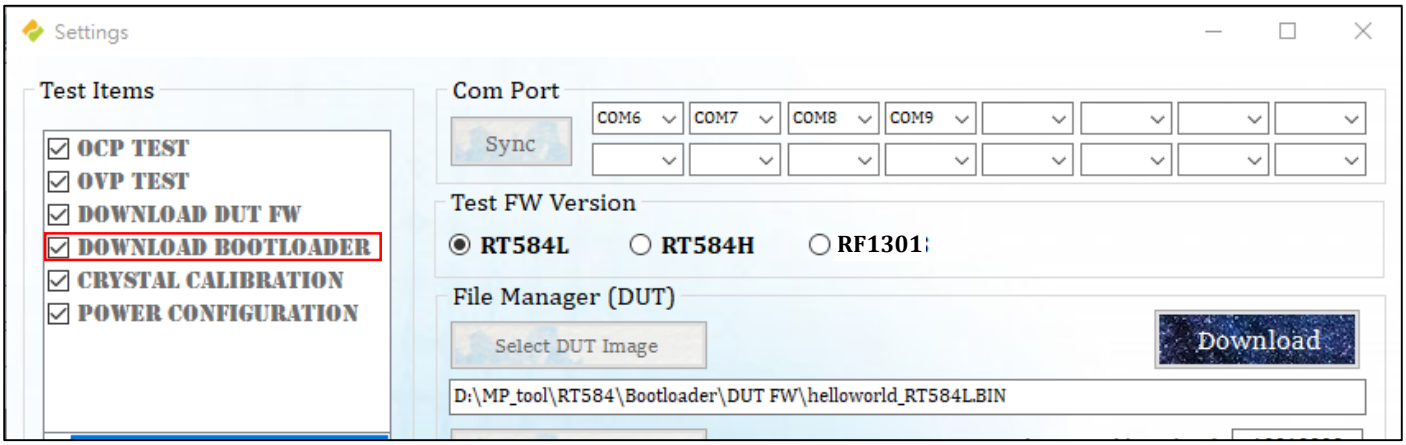


Figure 25. Check the test item - DOWNLOAD BOOTLOADER.

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Figure 26. Click the TEST button.

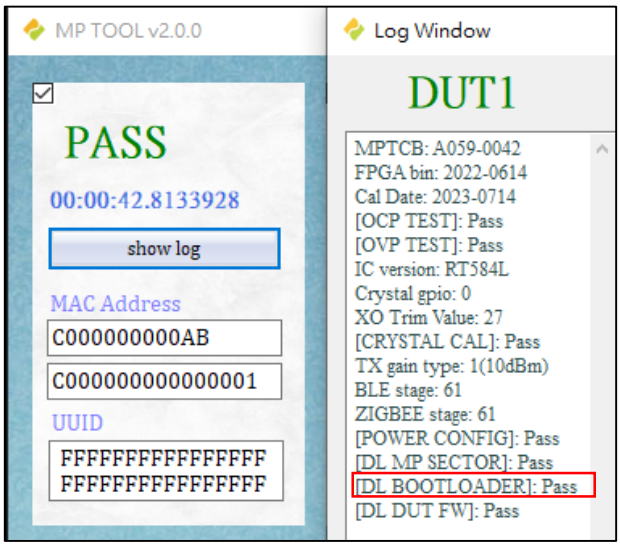


Figure 27. Show the log of the DOWNLOAD BOOTLOADER.

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3.6 Download MP sector Firmware to the DUT

The MP sector binary is not provided for users to select. When the user performs testing, the MP Tool will automatically download the MP sector firmware to the DUT after all test items have been completed, and then write the MP sector calibration data into the DUT.

The procedure for downloading mp sector firmware to the DUT in the MP Tool is described below: (See Figure 28~29).

1. Go to the MP TOOL v2.0.0 page and press the TEST.
2. Press show log to view the result.

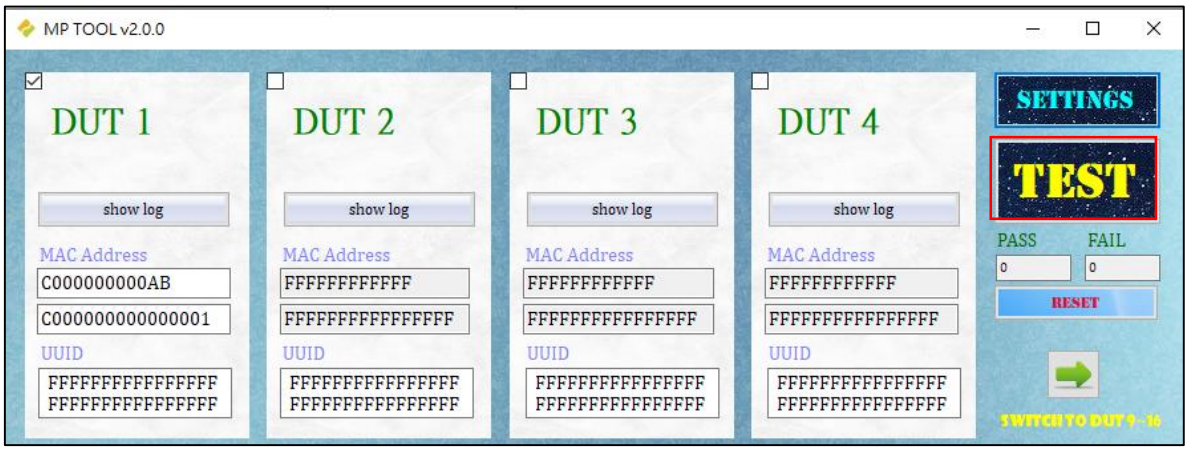


Figure 28. Click the TEST button.

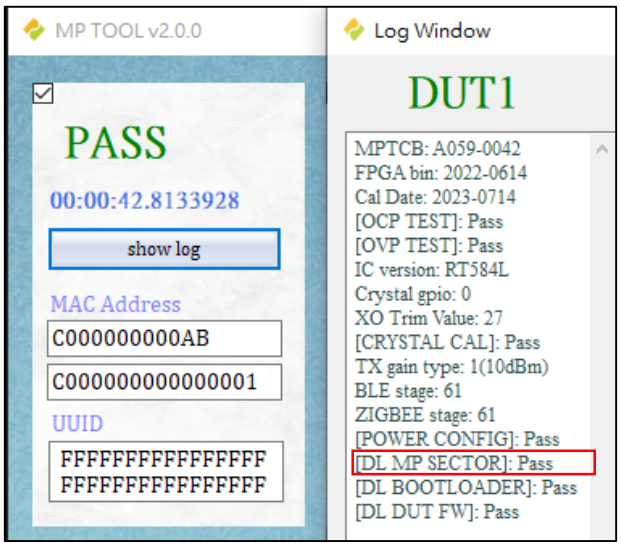


Figure 29. Show the log of the DOWNLOAD MP SECTOR.

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3.7 Crystal Calibration

The **Crystal Trim** test adjusts the crystal frequency offset by measuring the deviation of the crystal signal using the MPTCB.

The crystal calibration settings in the MP Tool are described below: (See Figure 30~33)

1. Go to the ConfigForm page and select the GPIO pin.
   - The DUT clock output options: Pin 0, 6, 20 or 21.
   - The Crystal trim input:
     i. RT584L/H/HA4: pin 2.
     ii. RF1301: pin 6.
2. Hardware Connection: On the test board, use a DuPont jumper wire to connect the **Crystal trim input pin** to the selected **DUT clock output pin**.
3. Go to the settings page and enable **CRYSTAL CALIBRATION**.
4. Go to the MP TOOL v2.0.0 page and press the **TEST**.
5. Press **show log** to view the result.

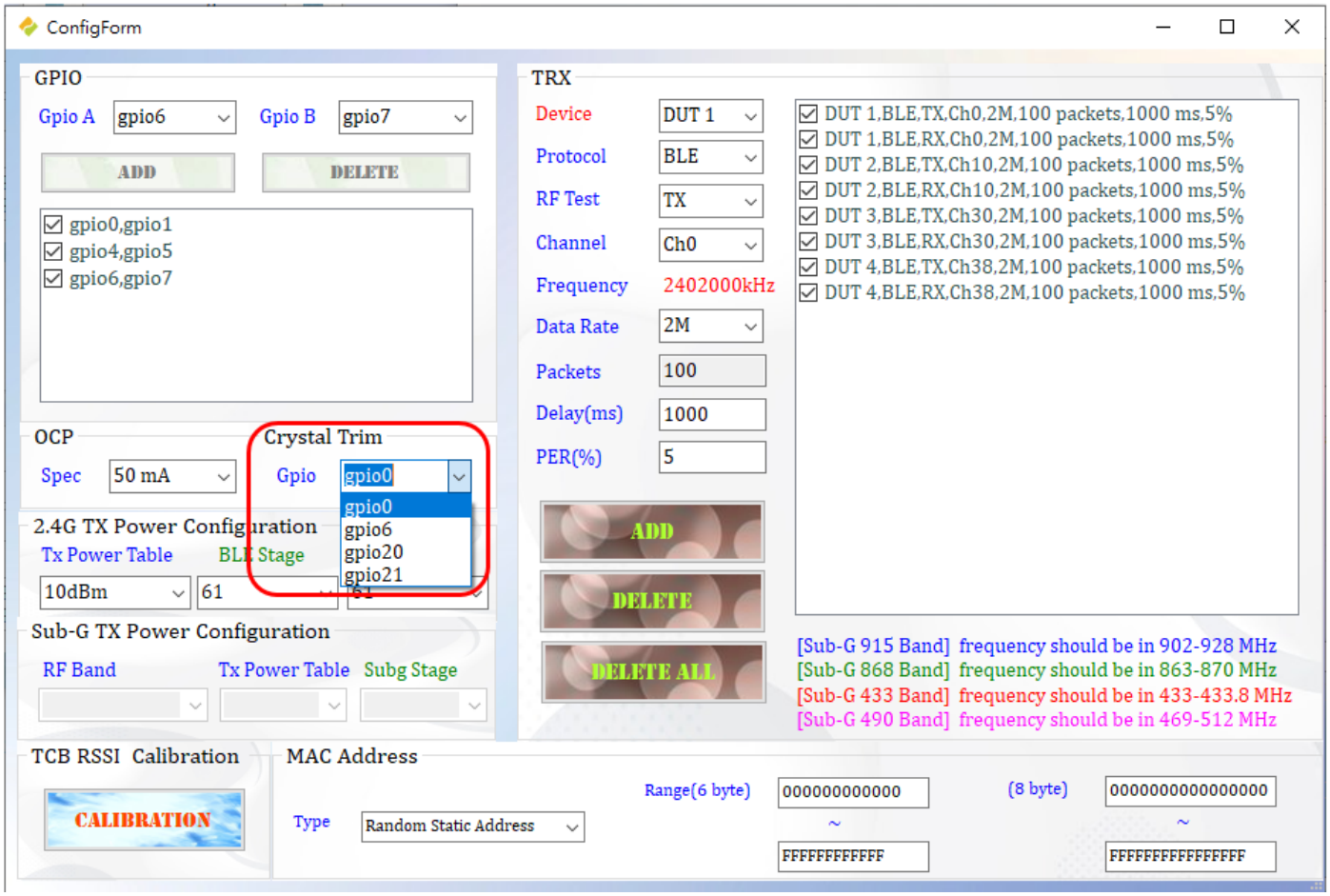


Figure 30. crystal trim settings

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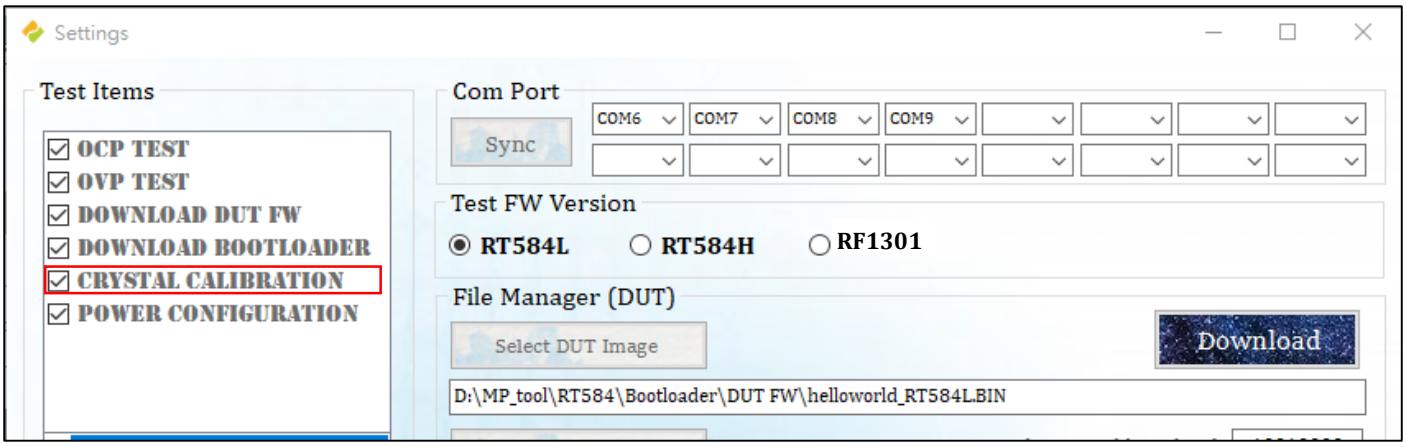


Figure 31. Check the test item - CRYSTAL CALIBRATION.

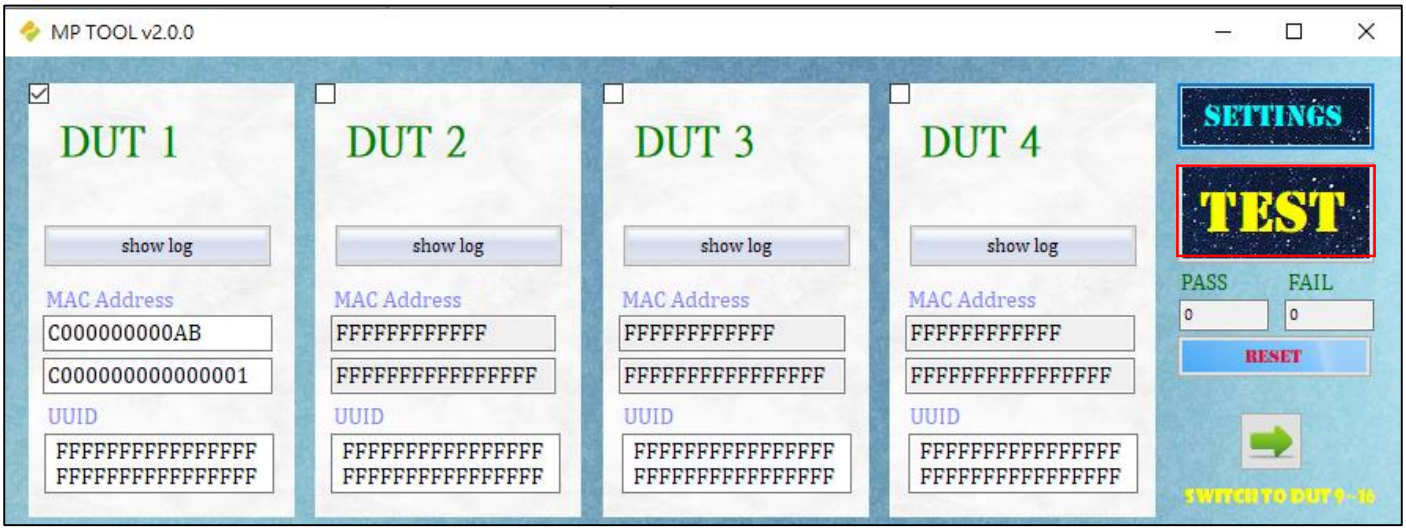


Figure 32. Click the TEST button.

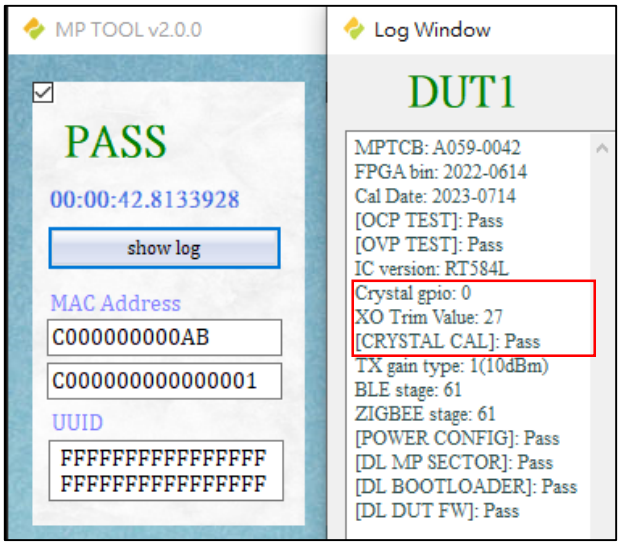


Figure 33. Show the log of the DOWNLOAD BOOTLOADER.

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3.8 Power Configuration

Set the RF TX power value for the chip's data transmission.

The power configuration in the MP Tool are described below: (See Figure 34~37)

1. Go to Settings page, select the Test FW version and enable **POWER CONFIGURAION**.
   - The TEST FW version is related the display of the power configuration on ConfigForm page.
   - The RT584L/ H only support 2.4GHz tx power.
     - RT584L support tx power: 0dBm and 10dBm
     - RT584H support tx power: 0dBm and 20dBm.
   - The RF1301 only support Sub-G tx power.
     - RF band is 902~928MHz/ 863~870MHz: 14dBm and 20dBm.
     - RF band is 433~435MHz/ 469~512MHz: 14dBm and 17dBm.
2. Go to the ConfigForm page and set tx power configuration.
3. Go to the MP TOOL v2.0.0 page and press the **TEST**.
4. Press **show log** to view the result.

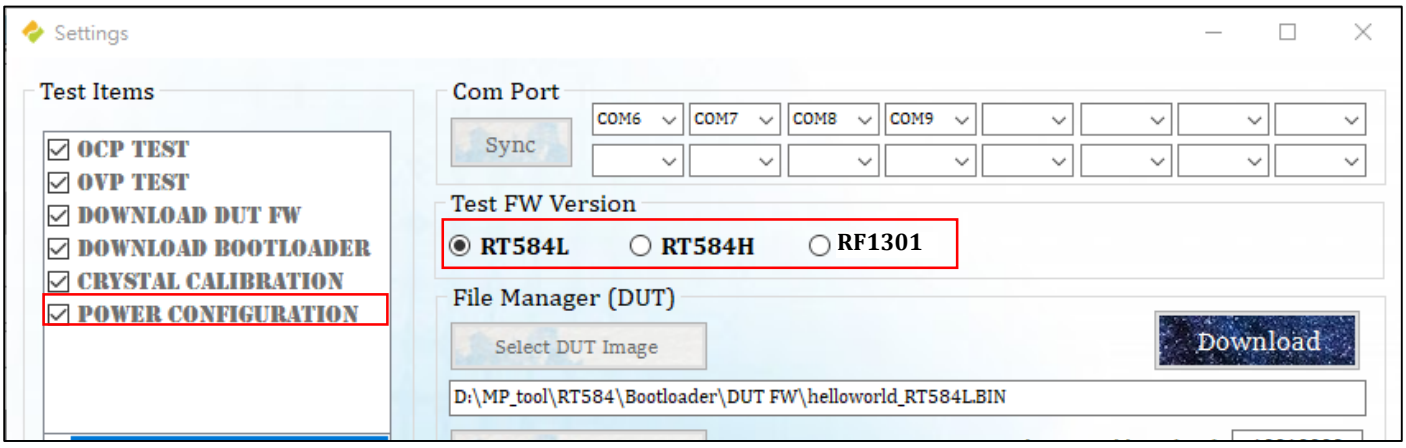


Figure 34. Select test fw version and check the test item - POWER CONFIGURAITON.

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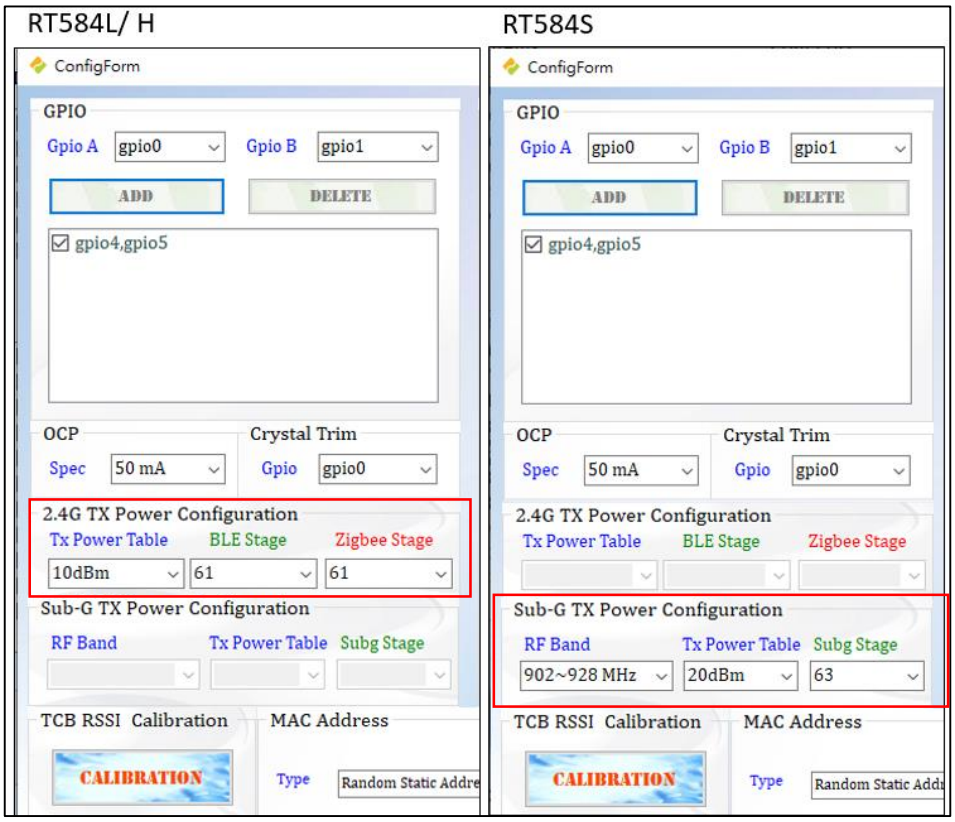


Figure 35. Set the tx power configuraion

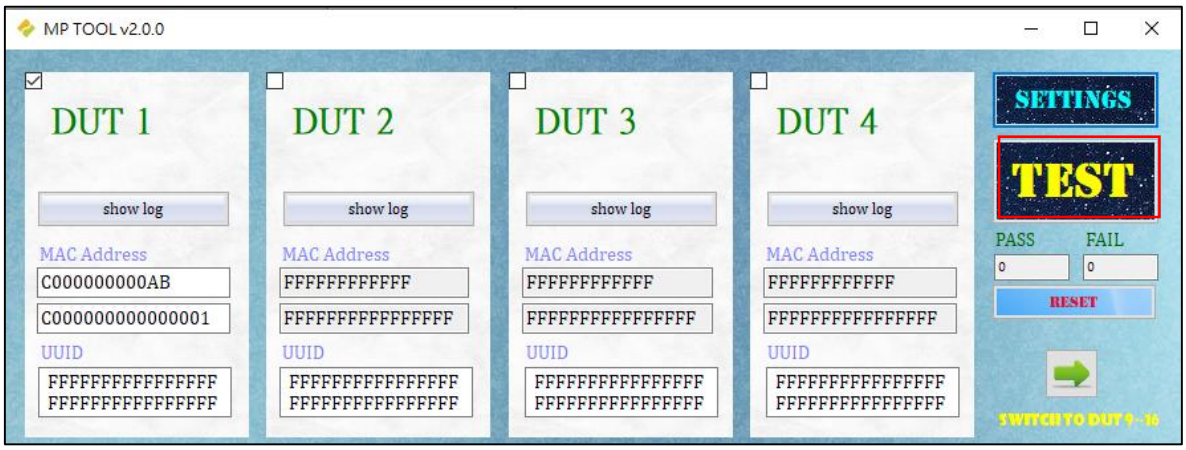


Figure 36. Click the TEST button.

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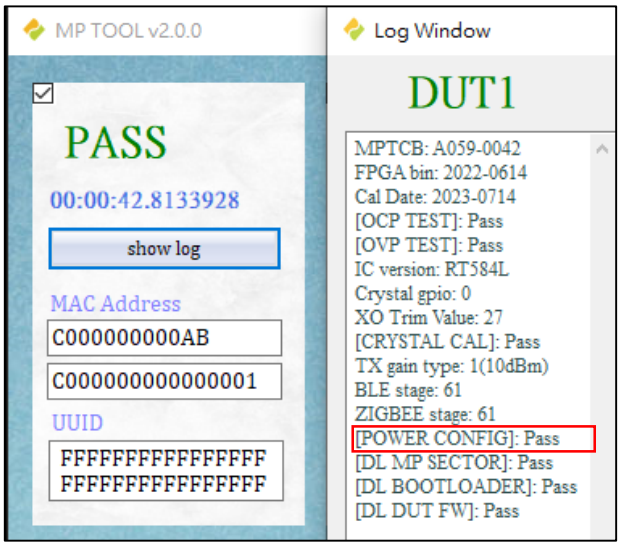


Figure 37. Show the log of the POWER CONFIGURATION

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3.9 GPIO Test

GPIO test is performed by verifying the connection between two GPIO pins.

The gpio test settings in the MP Tool are described below: (See Figure 38~41)

1. Go to the ConfigForm page and set GPIO pin pair.
   - Click **Add** in the **GPIO Test** section. The MP Tool will display the **GPIO Test Pair Dialog** window.
   - Click **DELETE** in the **GPIO Test** section. The MP Tool will disappear from the **GPIO Test Pair Dialog** window.
   - Select the GPIO pins you want to pair, then click the **pair** checkbox.
2. Go to Settings page and enable **GPIO TEST**.
3. Go to the MP TOOL v2.0.0 page and press the **TEST**.
4. Press **show log** to view the result.

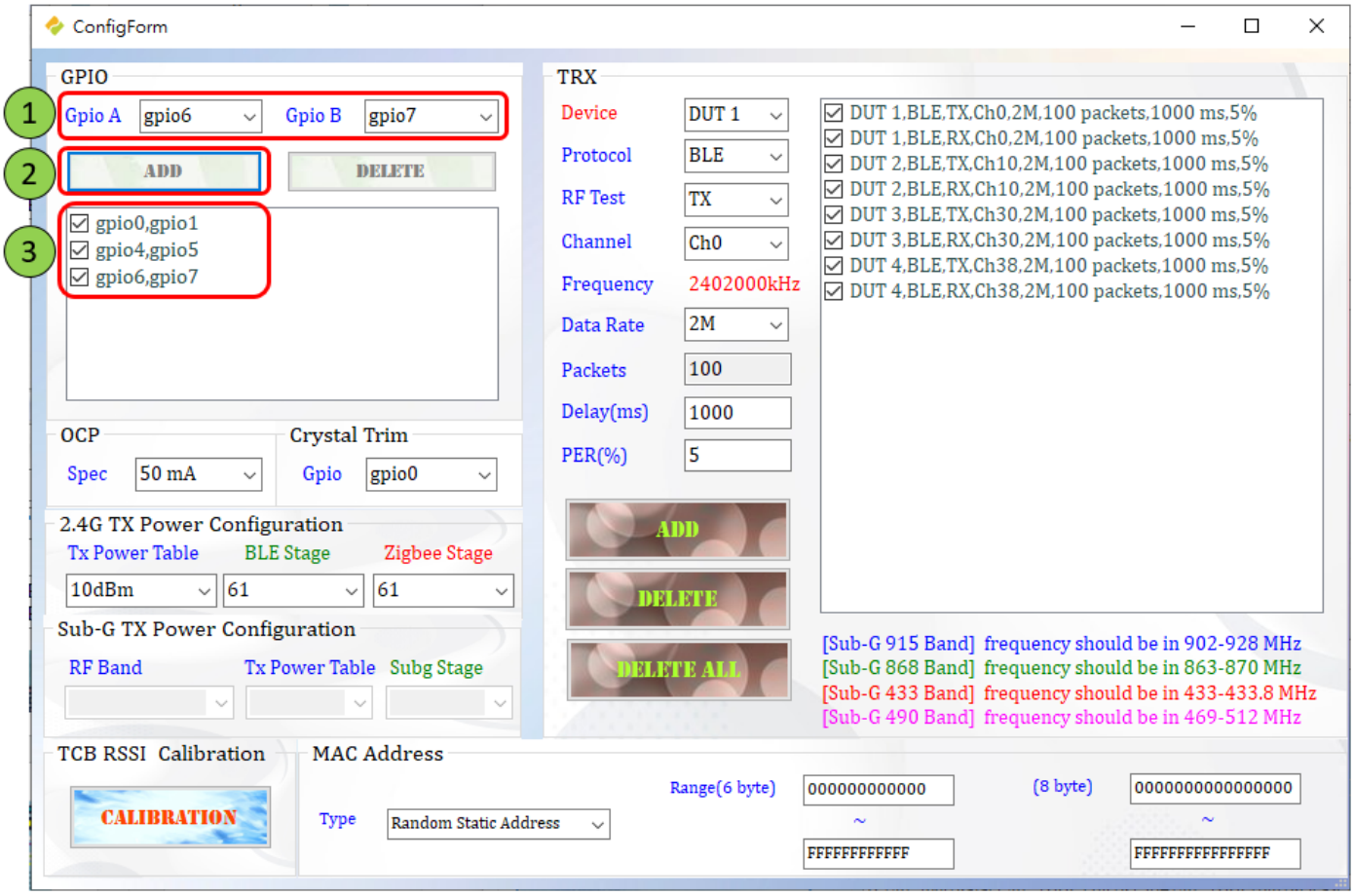


Figure 38. GPIO Test Setting

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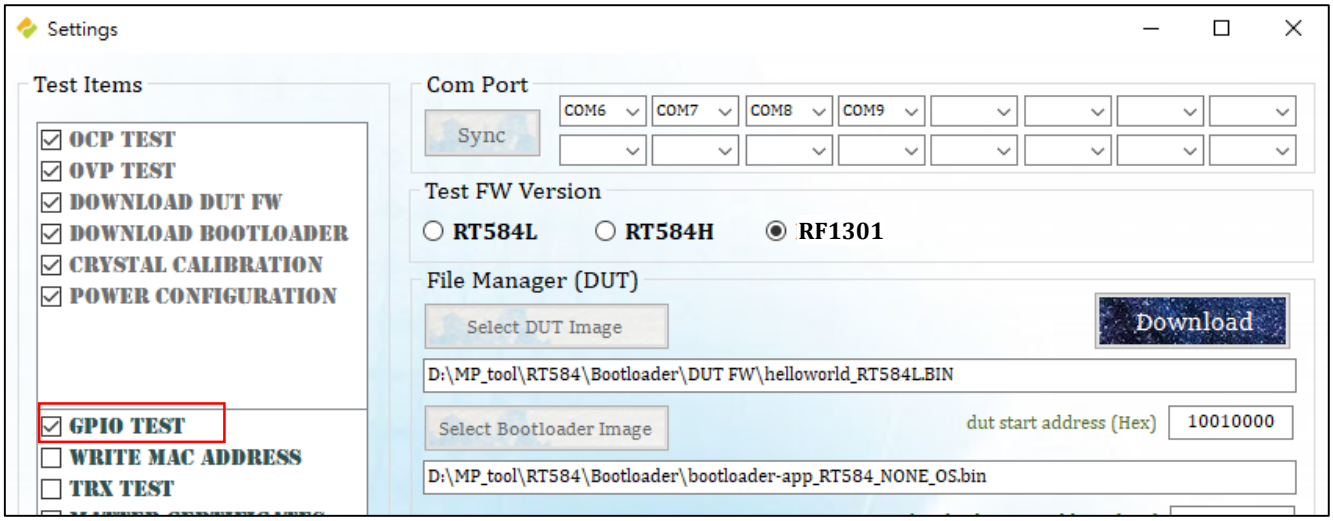


Figure 39. check the test item - GPIO TEST

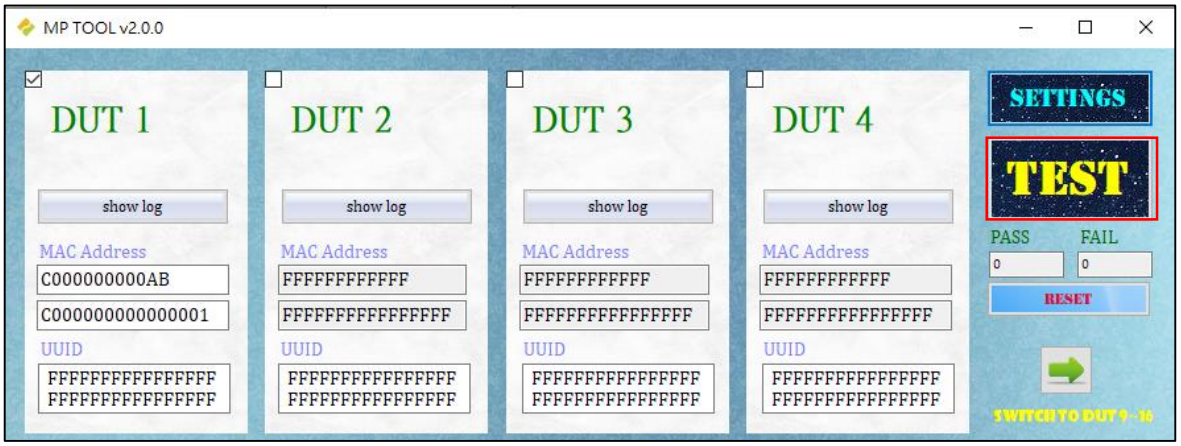


Figure 40. Click the TEST button.

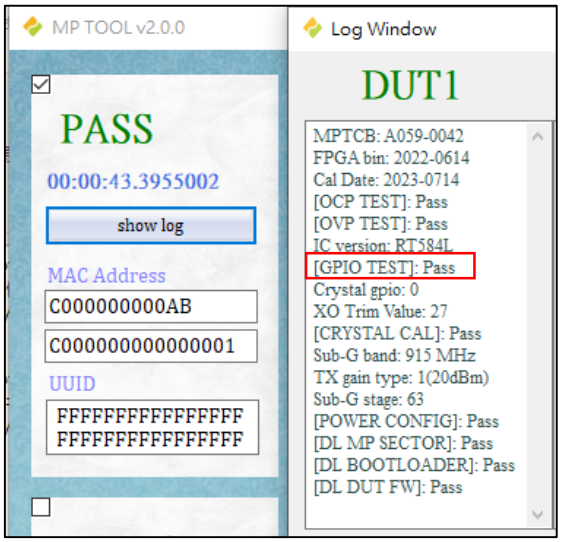


Figure 41.Show the log of the GPIO TEST.

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3.10 Write MAC Address

The MAC address settings in the MP Tool are described below: (See Figure 42~47)

1. Go to ConfigForm and select the **MAC address type**.
   - The default MAC address type is **Random Static Address**.
2. Set the available address range of BLE (6 bytes).
   - The range of Random Static Address is:
     - BLE: 0xC00000000000~0xCFFFFFFFFFFF.
   - The range of Public Address:
     - BLE: 0x000000000000~0xFFFFFFFFFFFF.
3. Set the available address range of range of Zigbee (8 bytes).
   - The default MAC address type is **Random Static Address**.
   - The range of Random Static Address is:
     - Zigbee: 0x0000000000000000~0xFFFFFFFFFFFFFFFF
   - The range of Public Address:
     - Zigbee: 0x0000000000000000~0xFFFFFFFFFFFFFFFF
4. Go to the settings page and enable **WRITE MAC ADDRESS**
5. Go to the MP TOOL v2.0.0 page, editing the start address for the first DUT and press the **TEST**.
6. Press the **show log** button to view the result.
   - After the mac address is written successfully, the tool will automatically increment the MAC address by one for the next DUT.
7. Verify the MAC address from the DUT.
   - Press **READ MAC**, then click **show log** to display the MAC address read from the DUT.

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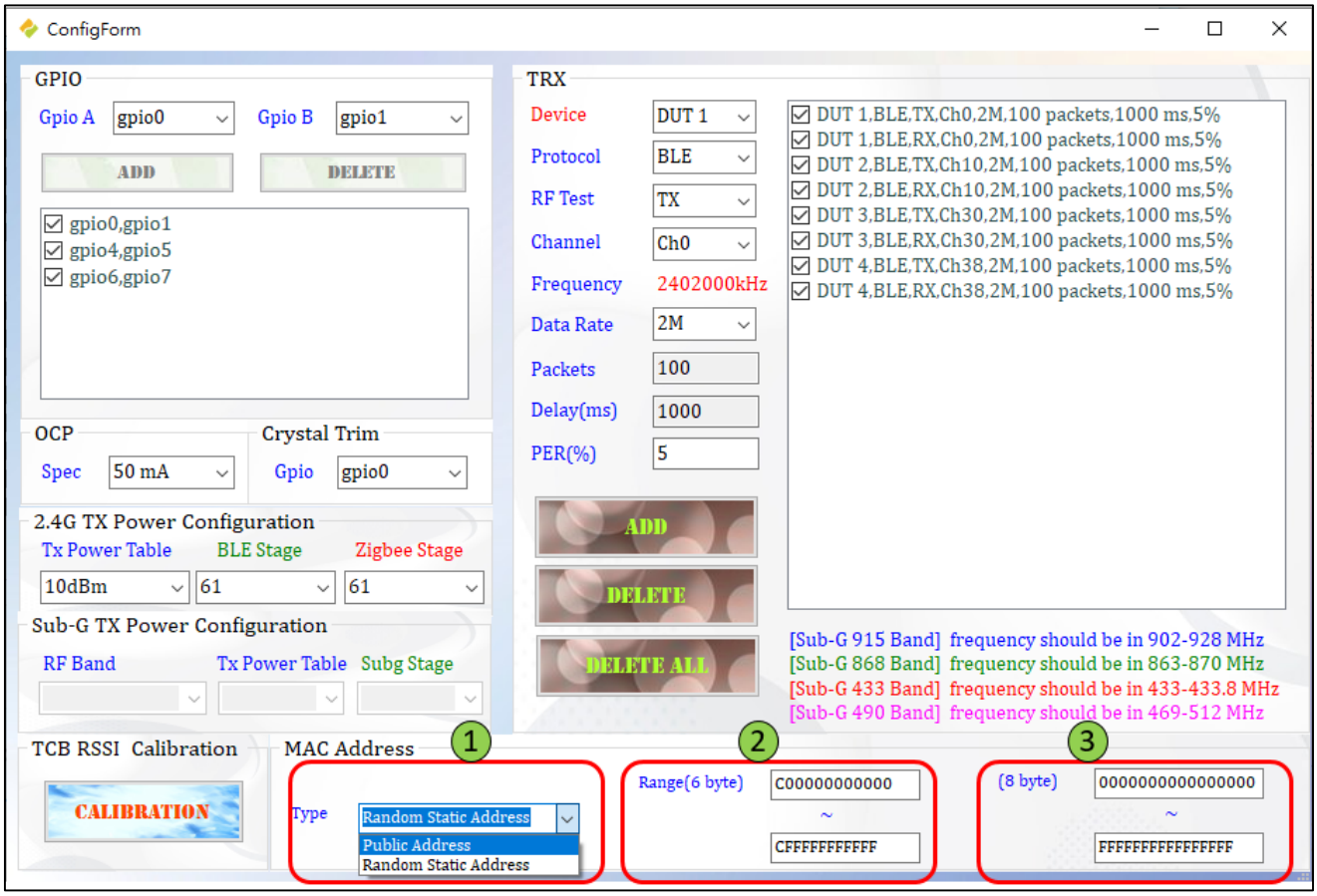


Figure 42. MAC Address Setting

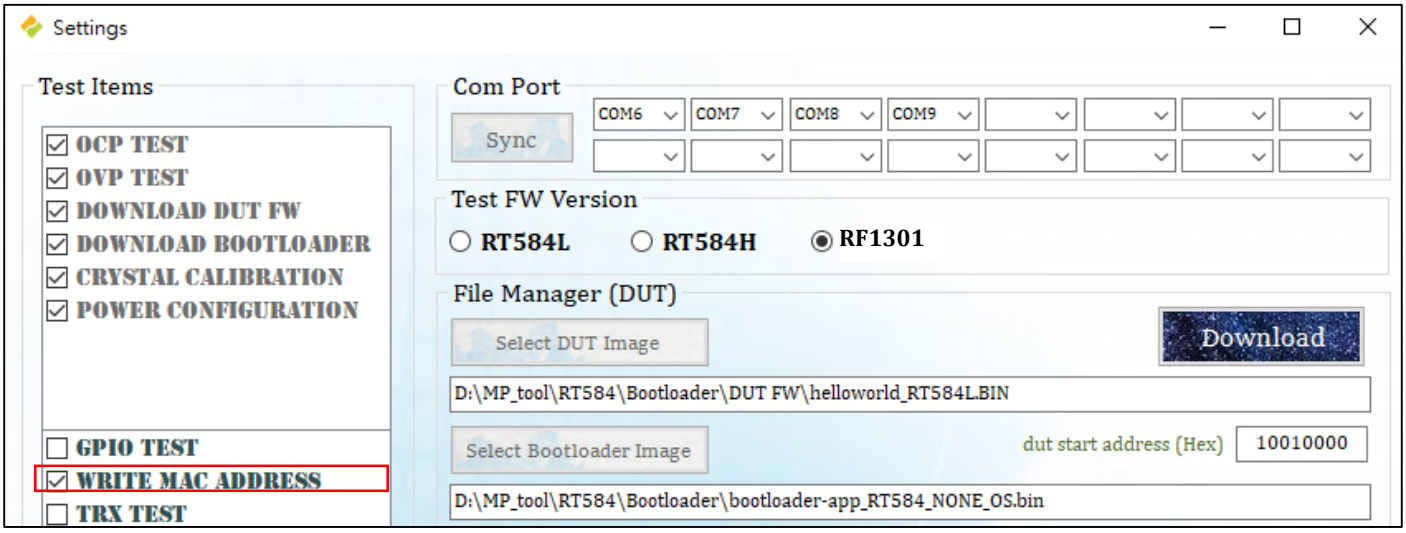


Figure 43. check the test item - WRITE MAC ADDRESS

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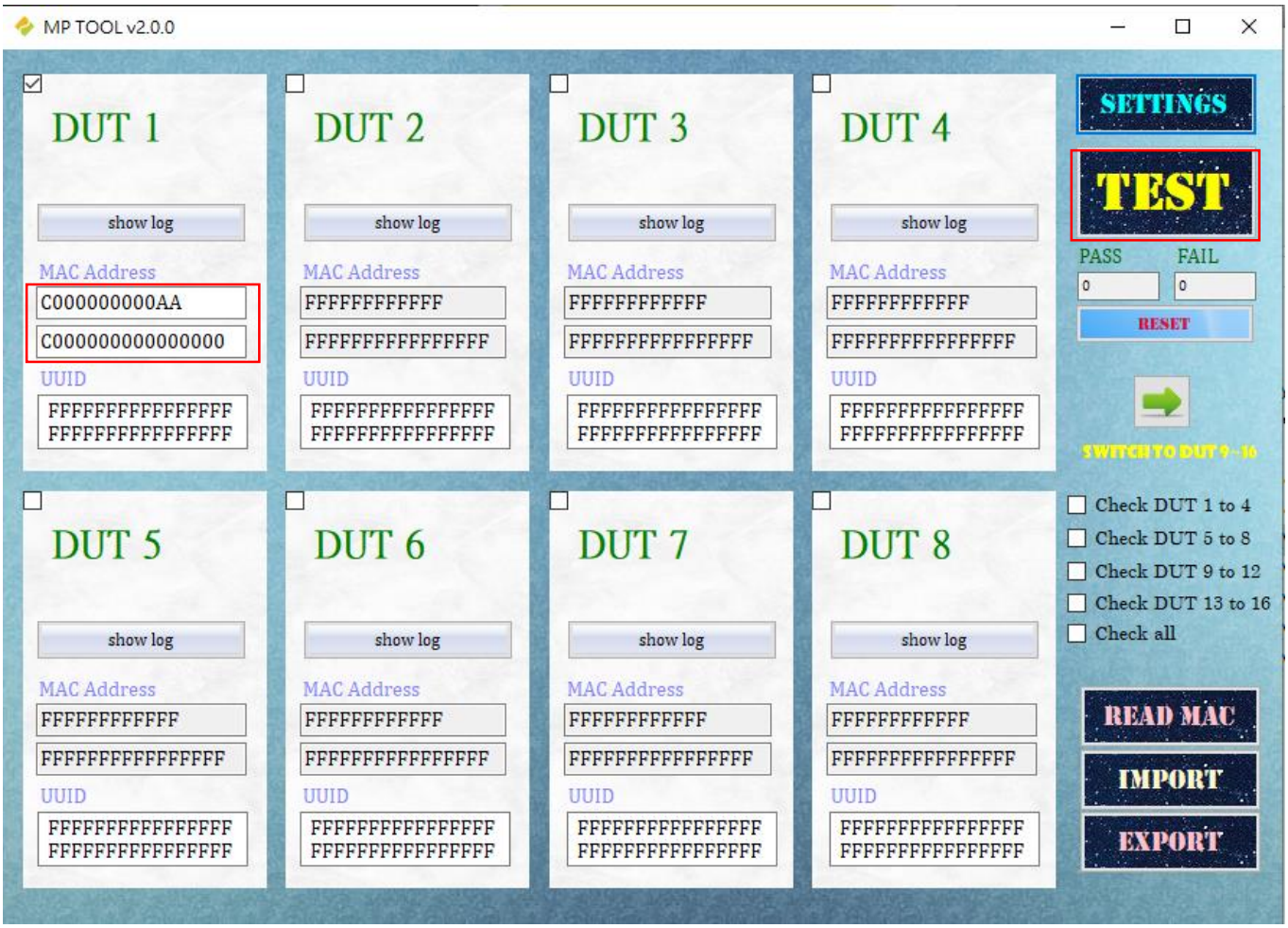


Figure 44. Edit the MAC address

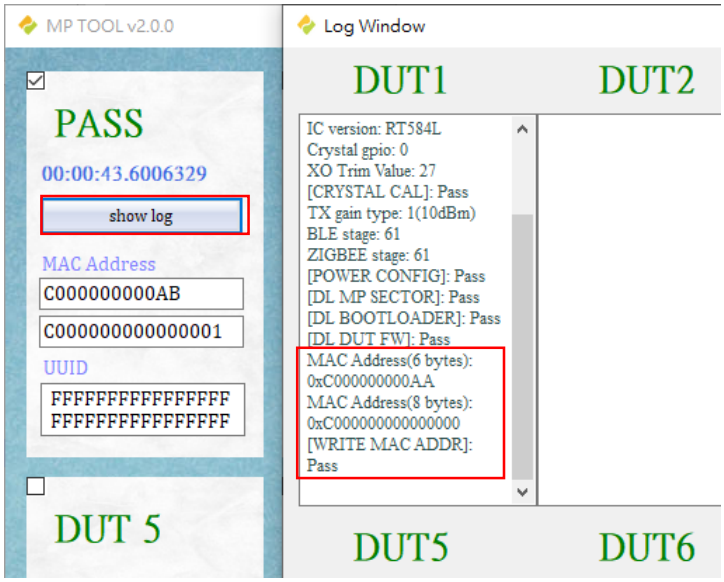


Figure 45. Show log of the MAC address

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Figure 46. Read the MAC address from DUT



Figure 47. Show log of MAC address from DUT

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3.11 RF TX/RX test

To test **Bluetooth, Zigbee** and **Sub-GHz** TX/RX settings.

The DUT transmits TX burst signals on the specified channel, and the MPTCB RX measures the DUT's transmitted packets to check whether the Packet Error Rate (PER) is abnormal. If the PER is within acceptable range, the DUT then receives TX burst packets from the MPTCB and measures the PER accordingly.

The RT584L/H support BLE and Zigbee protocols, while the RF1301 supports only the sub-GHz protocol.

The trx test settings in the MP Tool are described below: (See Figure 48~51)

1. Go to Settings page, select the Test FW version and enable **TRX TEST**.
   - The TEST FW version is related the display of the TRX default test item on ConfigForm page.
2. Go to the ConfigForm page and set trx test items.
   - The RT584L/ H only support 2.4GHz tx power. Please add the BLE/ Zigbee trx test items.
   - The RF1301 only support Sub-G tx power. Please add the Sub-G trx test items.
3. Back on the test board. **Using SMA male to SMA male RF cable connect DUT RF SMA connector and the corresponding dongle's RF SMA connector**.
4. Go to the MP TOOL v2.0.0 page and press the **TEST**.
5. Press **show log** to view the result.

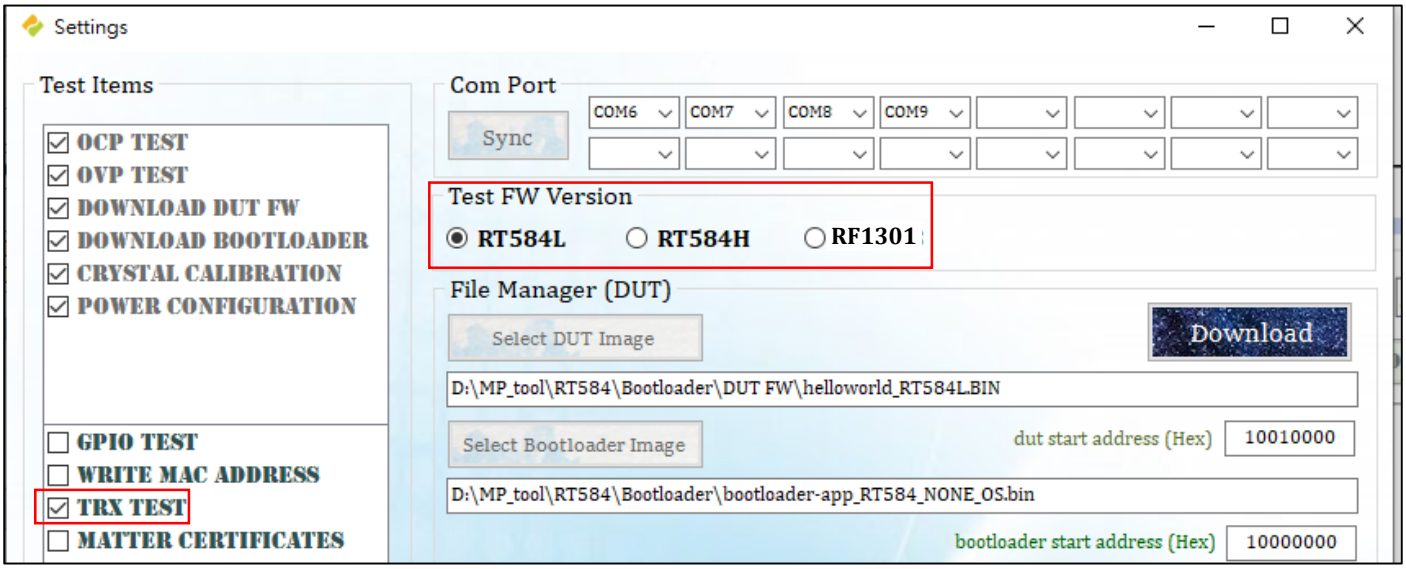


Figure 48. Select test fw version and check the test item - TRX TEST.

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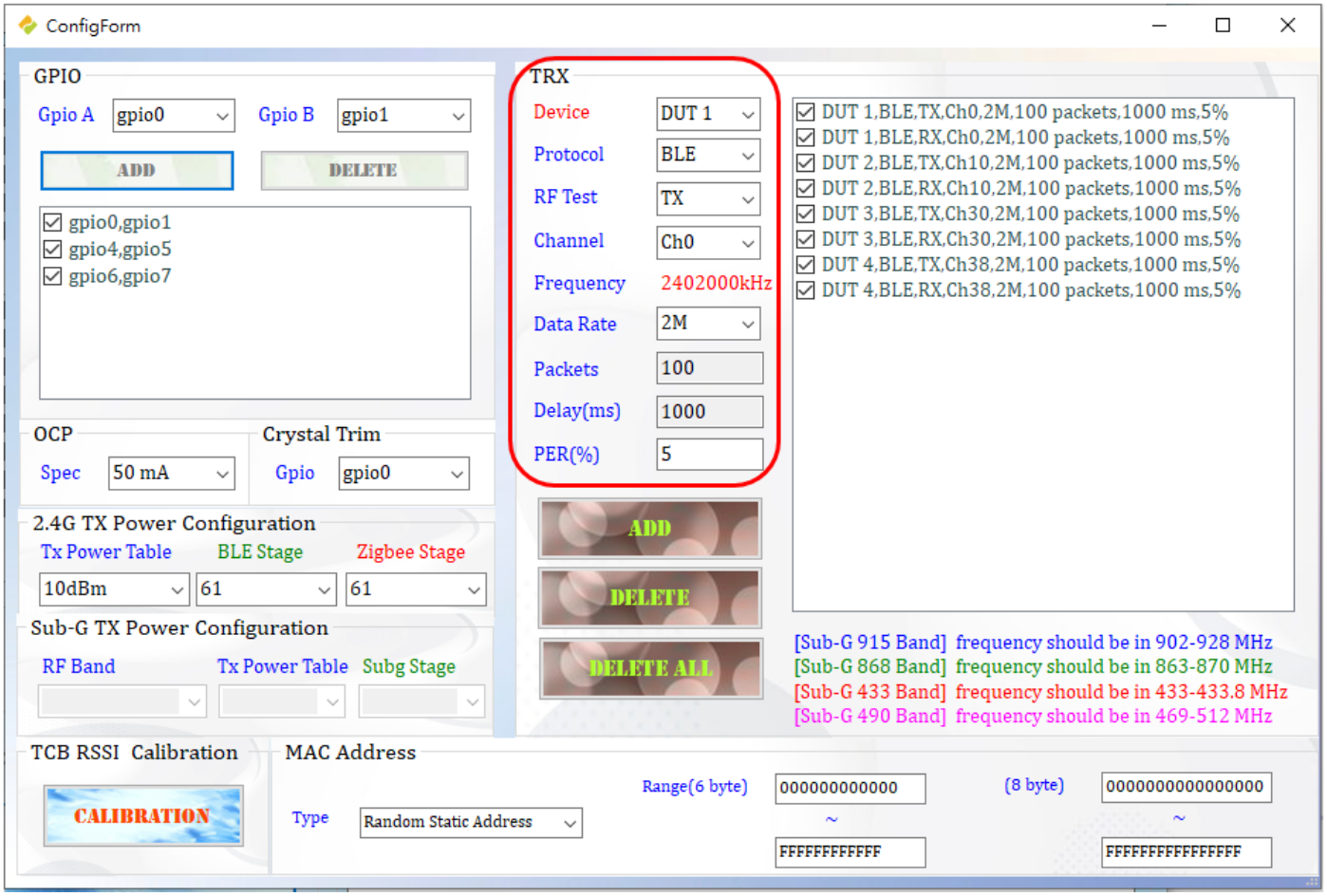


Figure 49. RF TX/RX Test Setting

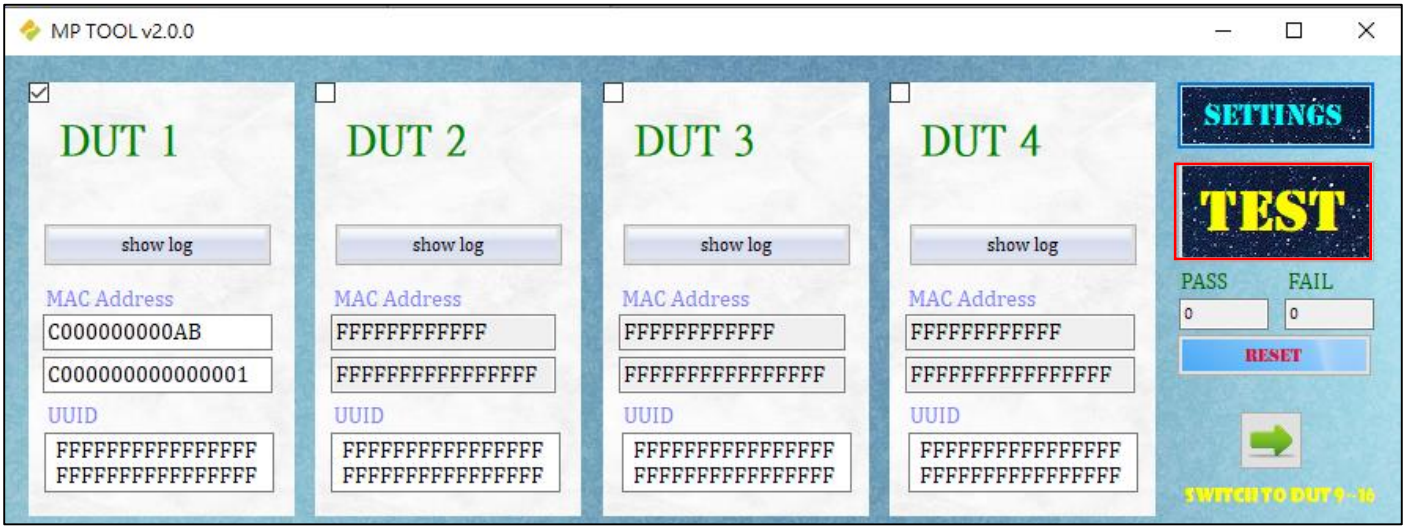


Figure 50. Click the TEST button.

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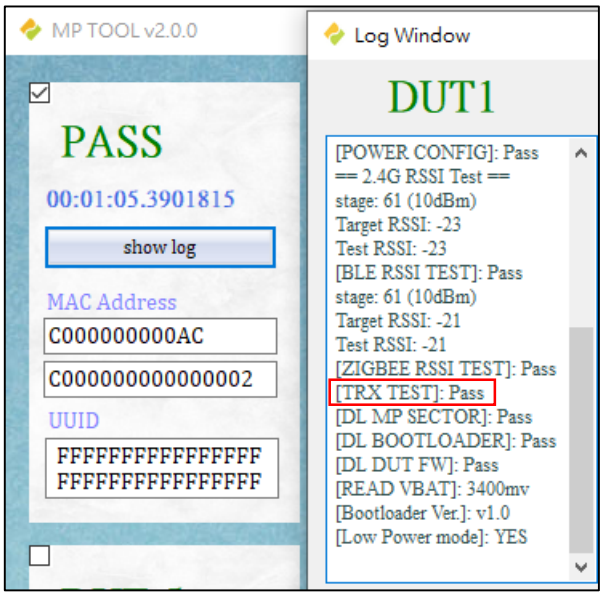


Figure 51. Show log of the TRX TEST

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3.12 Matter production with Certificates

The procedure for Matter production with certificates is outlined below: (See Figure 52~ 54)

1. Go to Settings page, select the certificate folder and enable **MATTER CERTIFICATES**.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

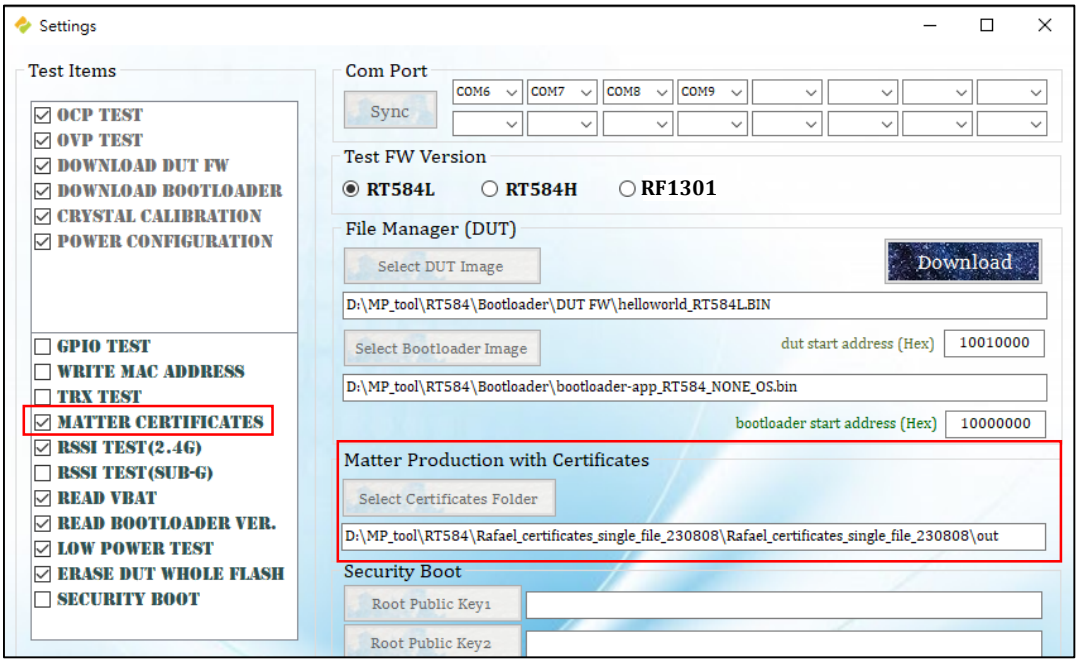


Figure 52. Select certificates folder

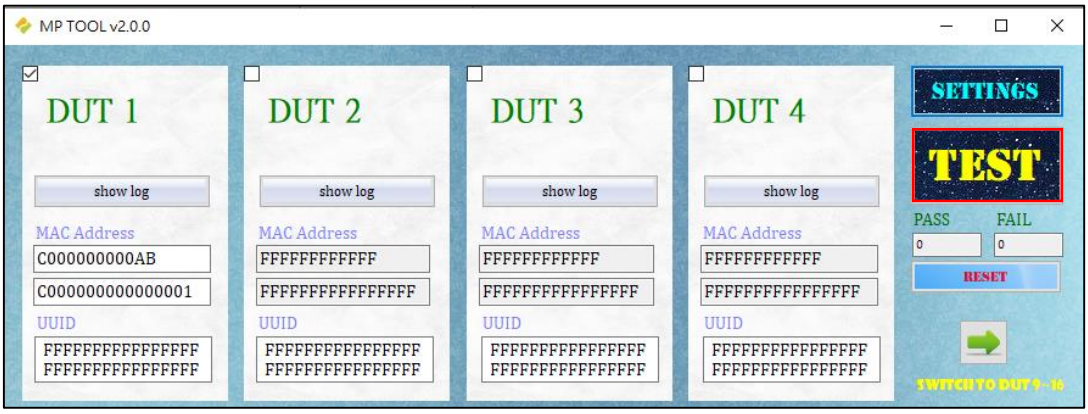


Figure 53. Click the TEST button.

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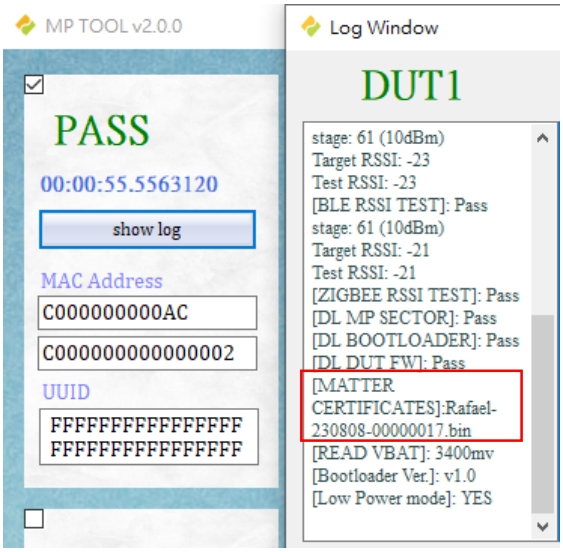


Figure 54. Show log of the MATTER CERTIFICATES.

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3.13 RSSI Test

Verify the DUT's Tx power.

The RSSI test settings in the MP Tool are described below: (See Figure 55~59)

1. Go to the Settings page, select the Test FW version, enable **RSSI TEST** and select **Cable Mode**.
   - RT584L/ H only support 2.4GHz; please enable **RSSIT(2.4G)**.
   - RF1301 only support Sub-G; please enable **RSSI(SUB-G)**.
2. Go to the ConfigForm page and set Tx power configuration.
   - Click the **CALIBRATION** button under the **TCB RSSI Calibration** section.
3. Go to the RSSI Calibration page
   - Check the COM port status. If no COM port is detected, click **Sync.**
   - Select Tx Cal Source:
     - **Golden DUT** - the DUT transmits a single tone.
     - **Single Generator** - a single generator transmits a single tone.
   - Adjust the stage value to change the output power.
   - If using a signal generator, ensure the output power is between **-10dBm** and **+10 dBm**.
4. On the test board, connect the DUT's RF SMA connector and the corresponding dongle's RF SMA connector using an **SMA male-to-SMA male RF cable**.
5. Confirm **2.4GHz/Sub-GHz path connection** between the MPTCB and the DUT, then click **Start Cal** to obtain the RSSI value.
6. Read the RSSI value from MPTCB and verify whether it matches the calibration value.
   - This RSSI value becomes the **Target RSSI**. Every DUT's RSSI test result will be compared against this value.
   - If **|RSSI – Target RSSI| > 5**, it indicates an issue with the DUT's Tx power, and the RSSI TEST will fail.
   - **Note:** The measured RSSI value is affected by the configured DUT Tx power, the attenuator value, and any surrounding interference.
7. Go to the MP TOOL v2.0.0 page and press the **TEST**.
8. Press **show log** to view the result.

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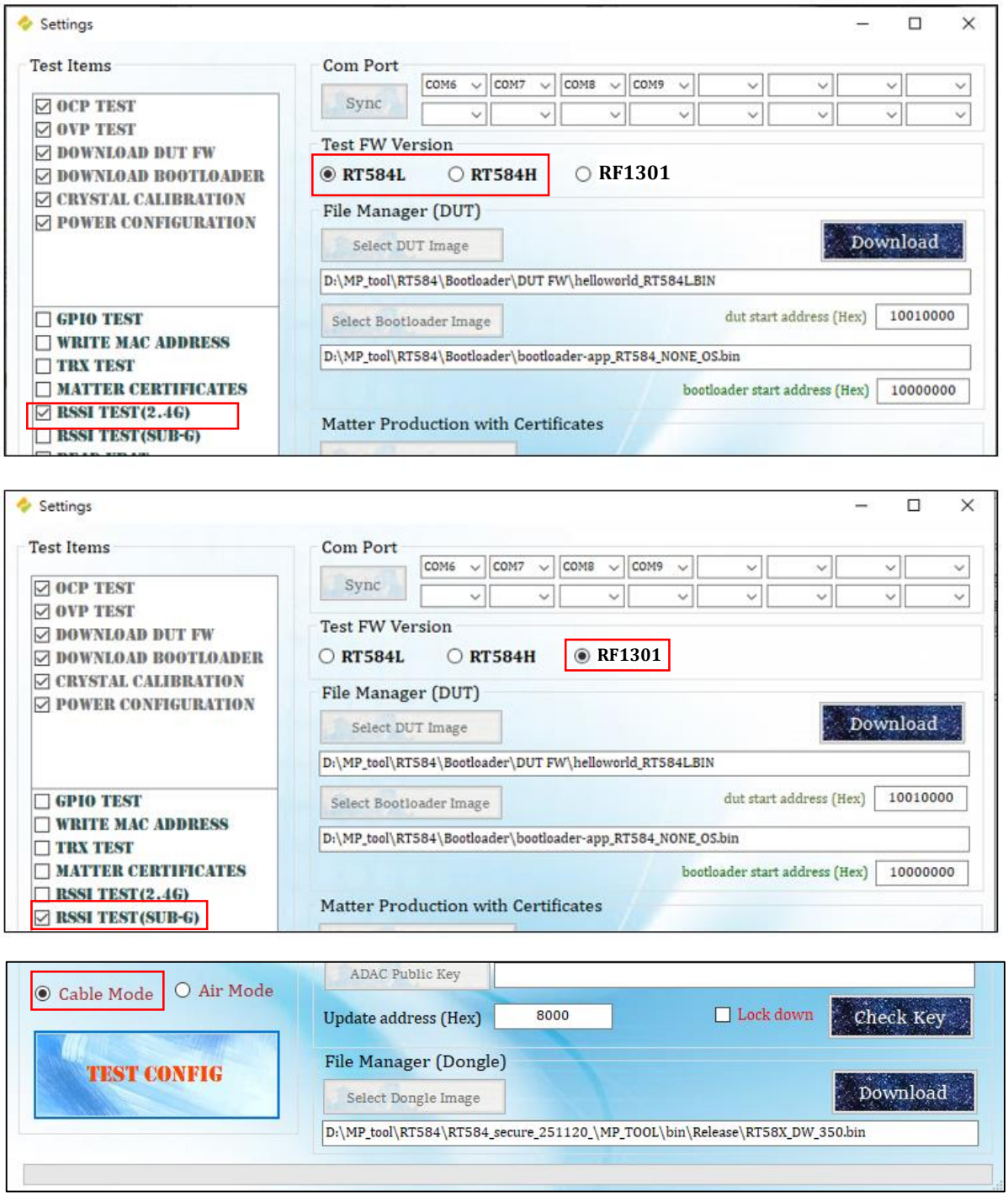


Figure 55. Select test fw version, cable mode and check the test item - RSSI TEST.

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Figure 56. Set the tx power configuraion

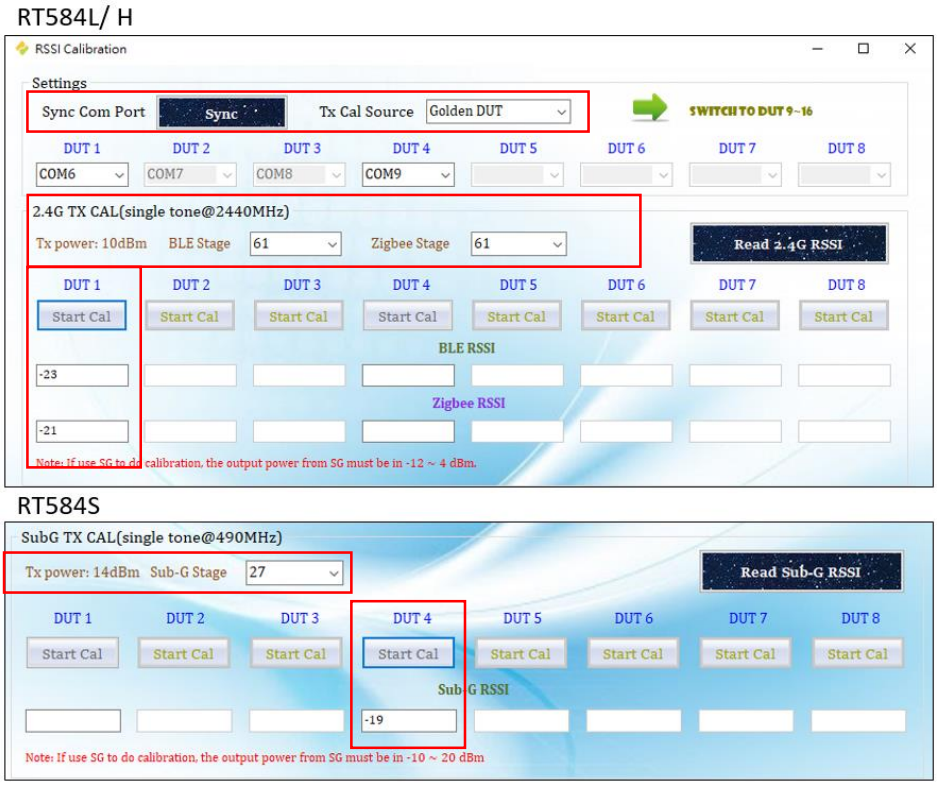


Figure 57. RSSI Calibration setting and detect DUT tx power.

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Note:

For RSSI calibration, users can estimate the approximate RSSI value using a simple calculation. To prevent high power input from damaging the chip, internal attenuators are built into both the 2.4GHz and Sub-GHz signal paths of the MPTCB.

Generally, the 2.4GHz path includes a 30 dB attenuator, while the Sub-GHz path includes a 40 dB attenuator.

For example, if the DUT transmits at 10dBm, the estimated RSSI would be:

2.4GHz path: 10dBm - 30dB = **-20dBm**

Sub-GHz path: 10dBm - 40db = **-30dBm**

Please note that the above values are theoretical estimates. Actual measured RSSI values are typically slightly lower than the theoretical values due to environmental attenuation and interference.

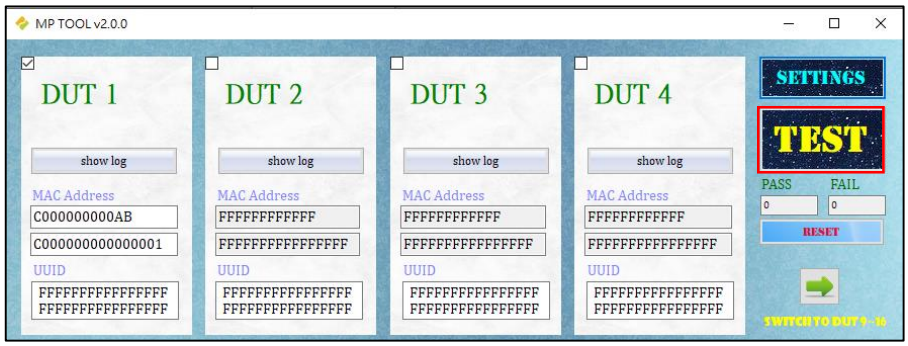


Figure 58. Click the TEST button.

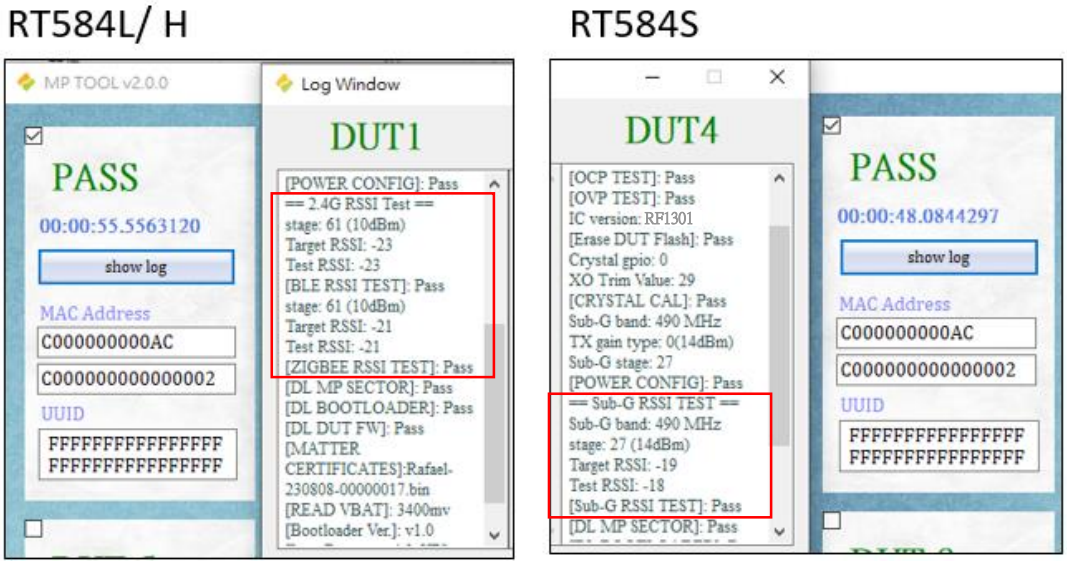


Figure 59. RSSI Test result

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3.14 Read VBAT

This test verifies whether the DUT can correctly read and report the battery voltage (VBAT) value through firmware or communication interface.

The read VBAT settings in the MP Tool are described below: (See Figure 60~62)

1. Go to Settings page and enable **READ VBAT**.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

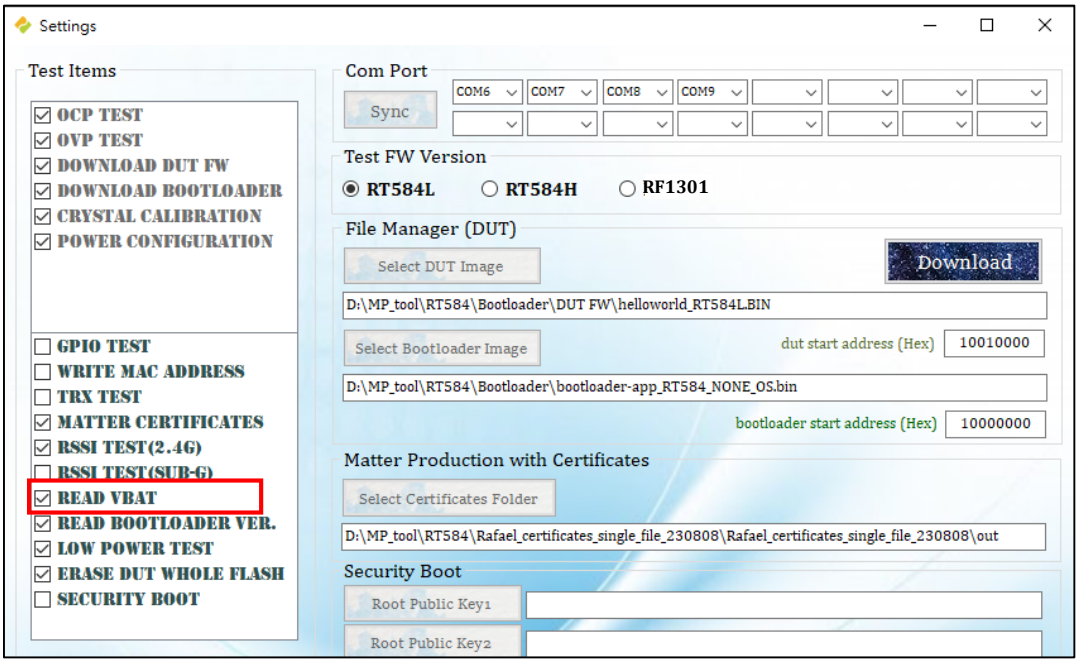


Figure 60. check the test item - READ VBAT.

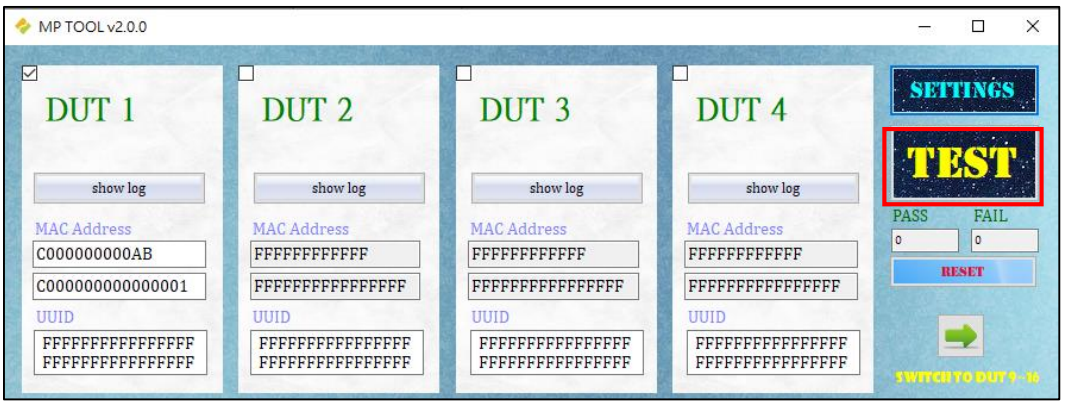


Figure 61. Click the TEST button.

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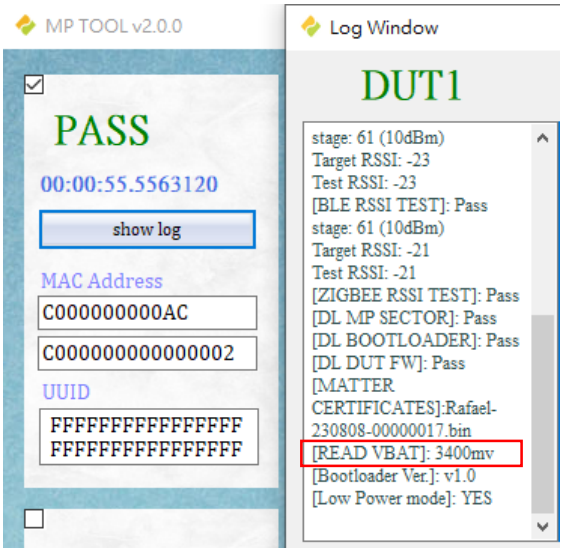


Figure 62. Show log of the READ VBAT.

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3.15 Read Bootloader Version

The steps to read the bootloader version in the MP Tool are described below: (See Figure 63~65)

1. Go to the Settings page and enable **READ BOOTLOADER VER**.
   - Please confirm that the DUT has the bootloader loaded.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

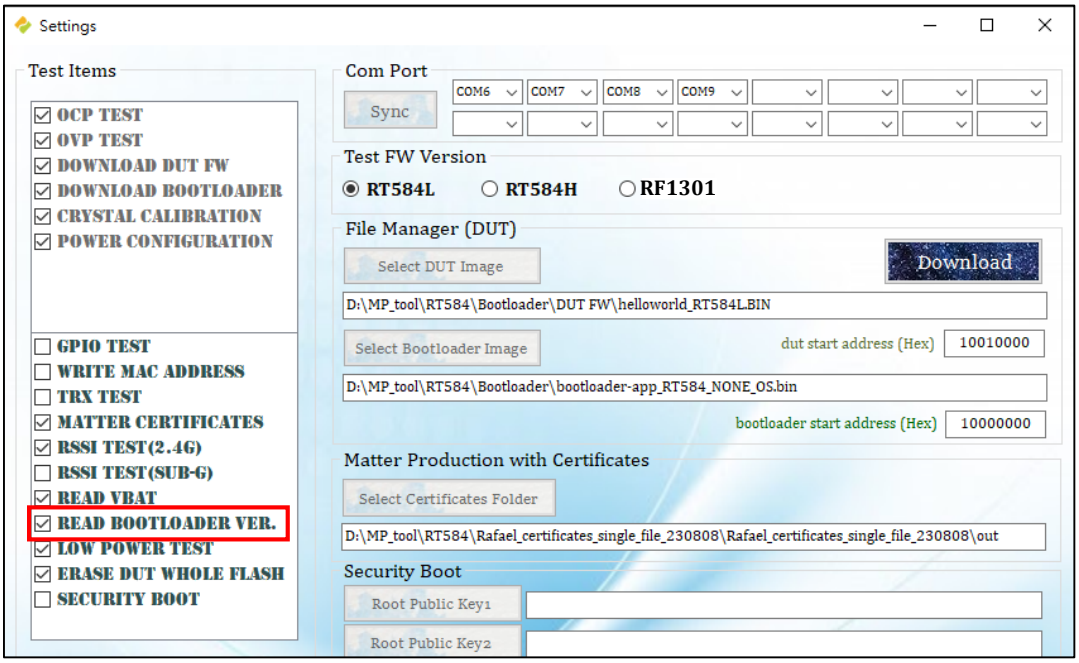


Figure 63. check the test item - READ BOOTLOADER VERSION.

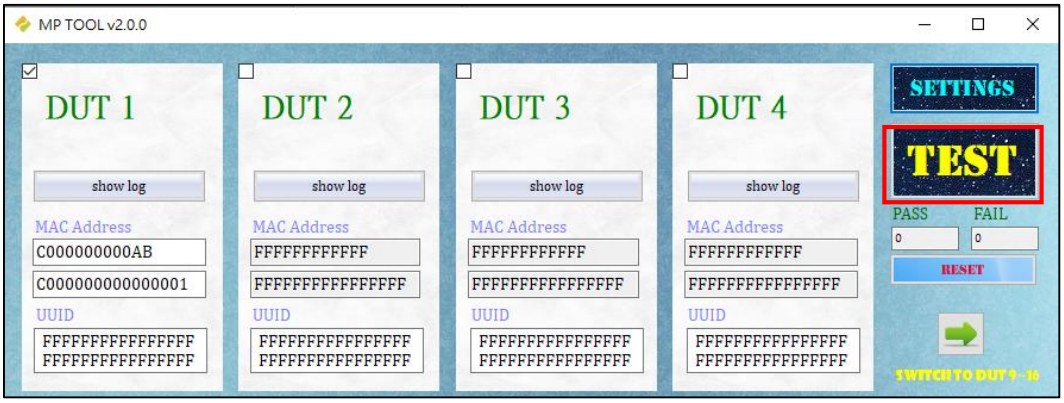


Figure 64. Click the TEST button.

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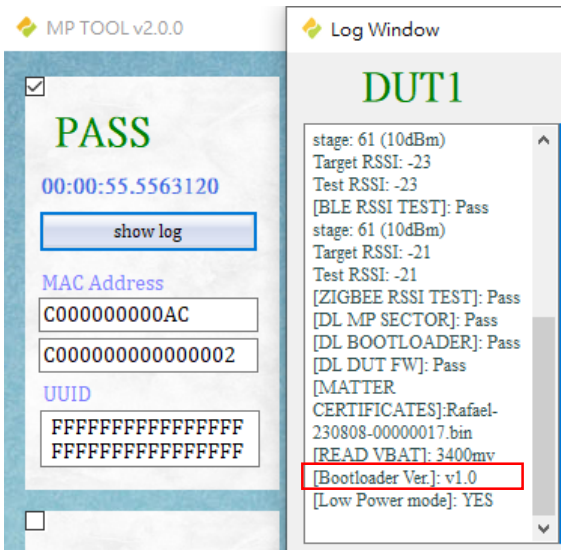


Figure 65. Show log of the READ BOOTLOADER VERSION

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3.16 Low Power Test

This test verifies that the DUT (Device Under Test) can enter low power mode correctly and meets the expected current consumption requirements. The MP Tool will send the command to let the DUT enter low power mode.

The steps to set DUT in low power mode in the MP Tool are described below: (See Figure 65~67)

1. Go to the Settings page and enable **LOW POWER TEST**.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

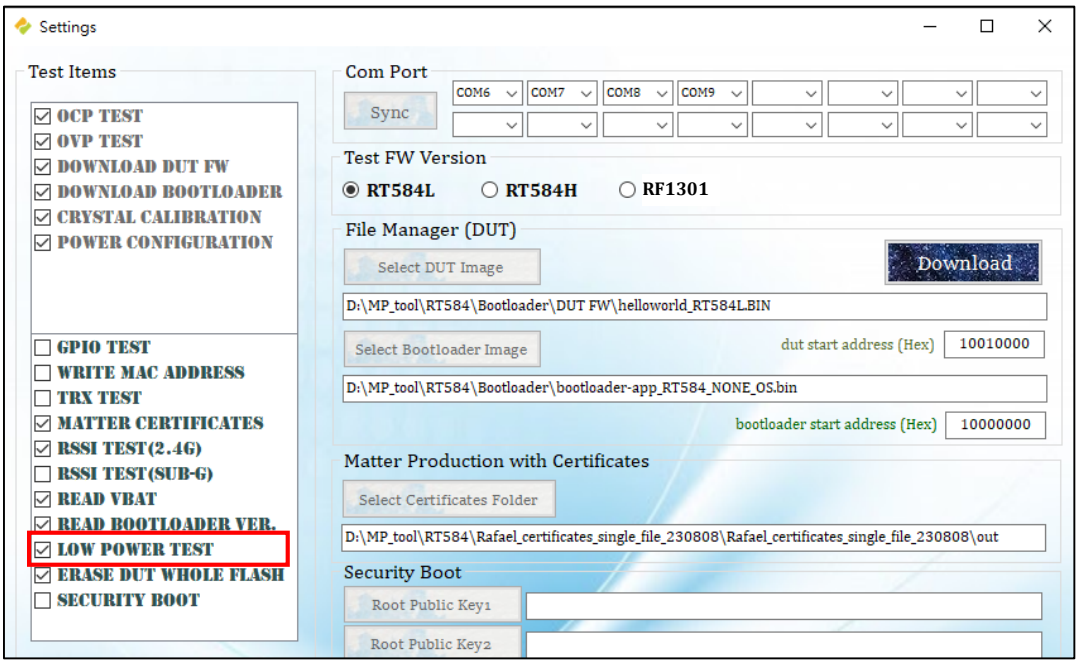


Figure 66. check the test item - LOAD POWER TEST.

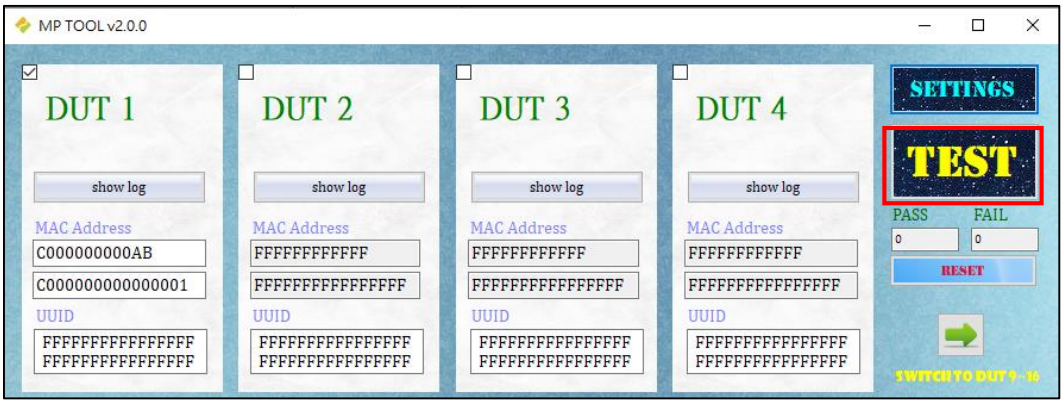


Figure 67. Click the TEST button.

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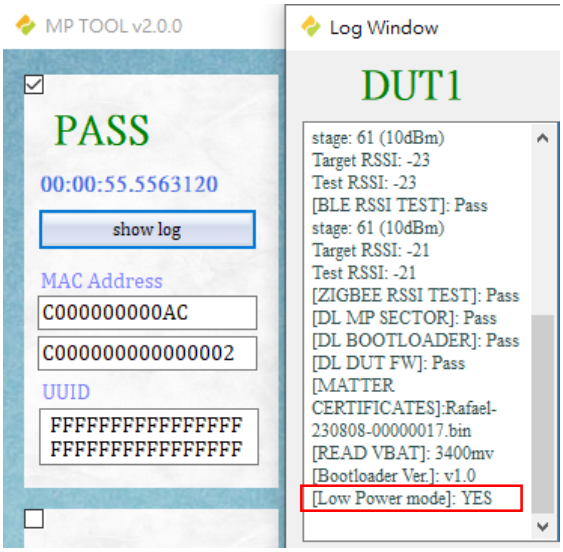


Figure 68. Show log of the LOW POWER TEST

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3.17 Erase DUT Whole Flash

It erases 2 MB of DUT flash for RT584L/H, or 1 MB for RF1301, starting from the bootloader address 0x10000000.

The steps to erase DUT whole flash in the MP Tool are described below: (See Figure 69~71)

1. Go to the Settings page and enable **ERASE DUT WHOLE FLASH**.
2. Go to the MP TOOL v2.0.0 page and press the **TEST**.
3. Press **show log** to view the result.

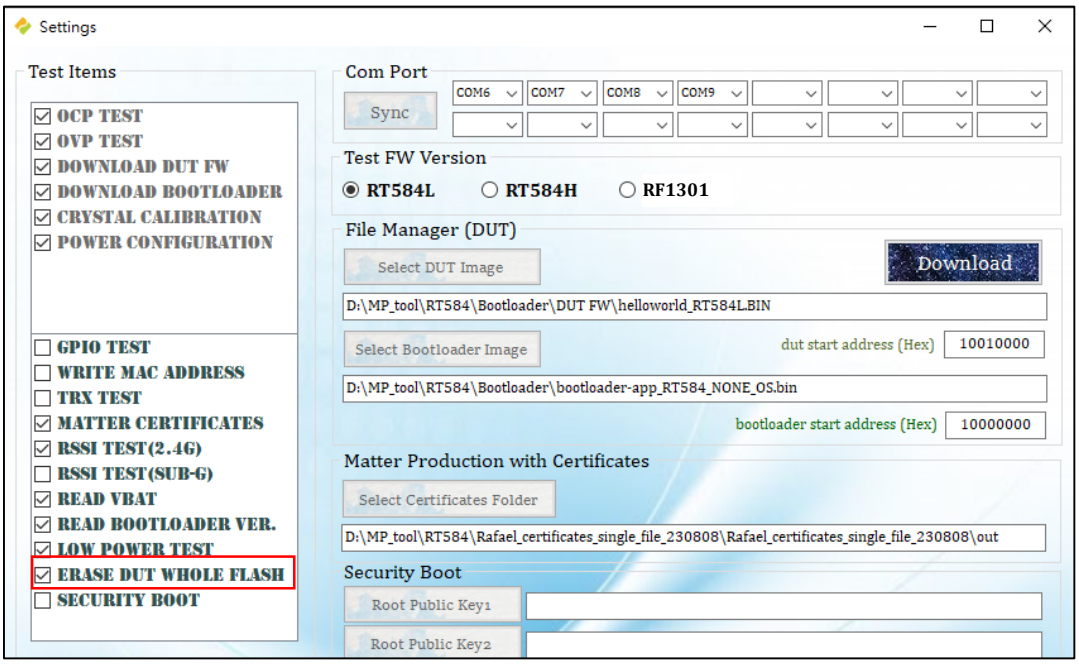


Figure 69. check the test item - ERASE DUT WHOLE FLASH.

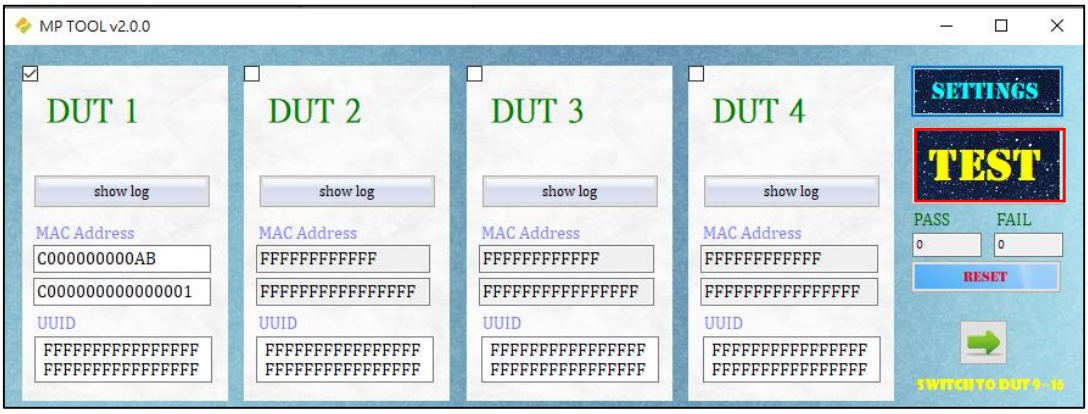


Figure 70. Click the TEST button.

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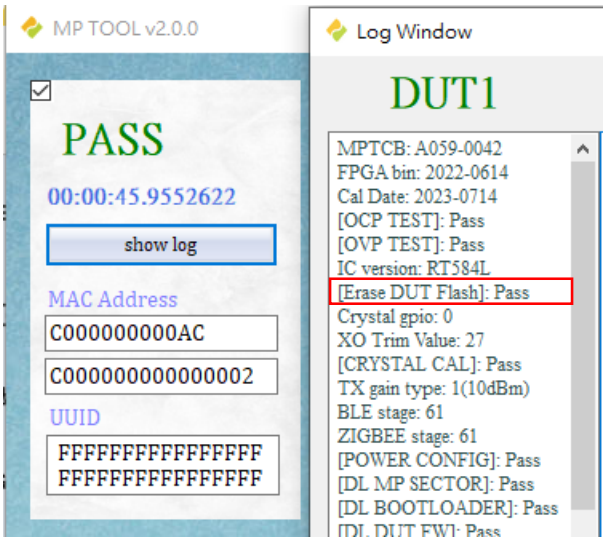


Figure 71. Show log of the ERASE DUT FLASH

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3.18 Secure Boot

If you want to load the secure bootloader, you must program both the public key and the ADAC key (for debug access) into the DUT; otherwise, the DUT will not boot successfully.

To confirm that the bootloader has not been modified, you can use the root public key and the signature public key to verify the bootloader.

However, the ADAC key cannot be verified by the bootloader, so please double-check it before writing it into the DUT.

Please note that once the public keys are written into the DUT, they cannot be changed.

The security boot settings in the MP Tool are described below: (See Figure 72~76)

1. Go to Settings page.
   - select the bootloader includes secured information.
   - Select the public keys.
     - Root public key 1, key 2 (optional), key 3 (optional)
     - Signature public key.
     - ADAC public key.
   - Enter the update address used for future bootloader updates.
     - Default value: 0x8000
   - Select the root public key that corresponds to the bootloader for verification.
   - Click **Verify Key**
2. **Enable SECURE BOOT** after successful verification of the root key and bootloader. This action locks the keys.
   - **Note:** If you need to modify keys or the bootloader, deselect **Lock Down**.
   - Re-locking requires repeating steps 1 and 2.
3. Select **Write Life Cycle to Security**. This sets the life cycle to **Security Mode** and permanently prevents further editing.
4. Go to the MP TOOL v2.0.0 page and press the **TEST**.
5. Press **show log** to view the result.

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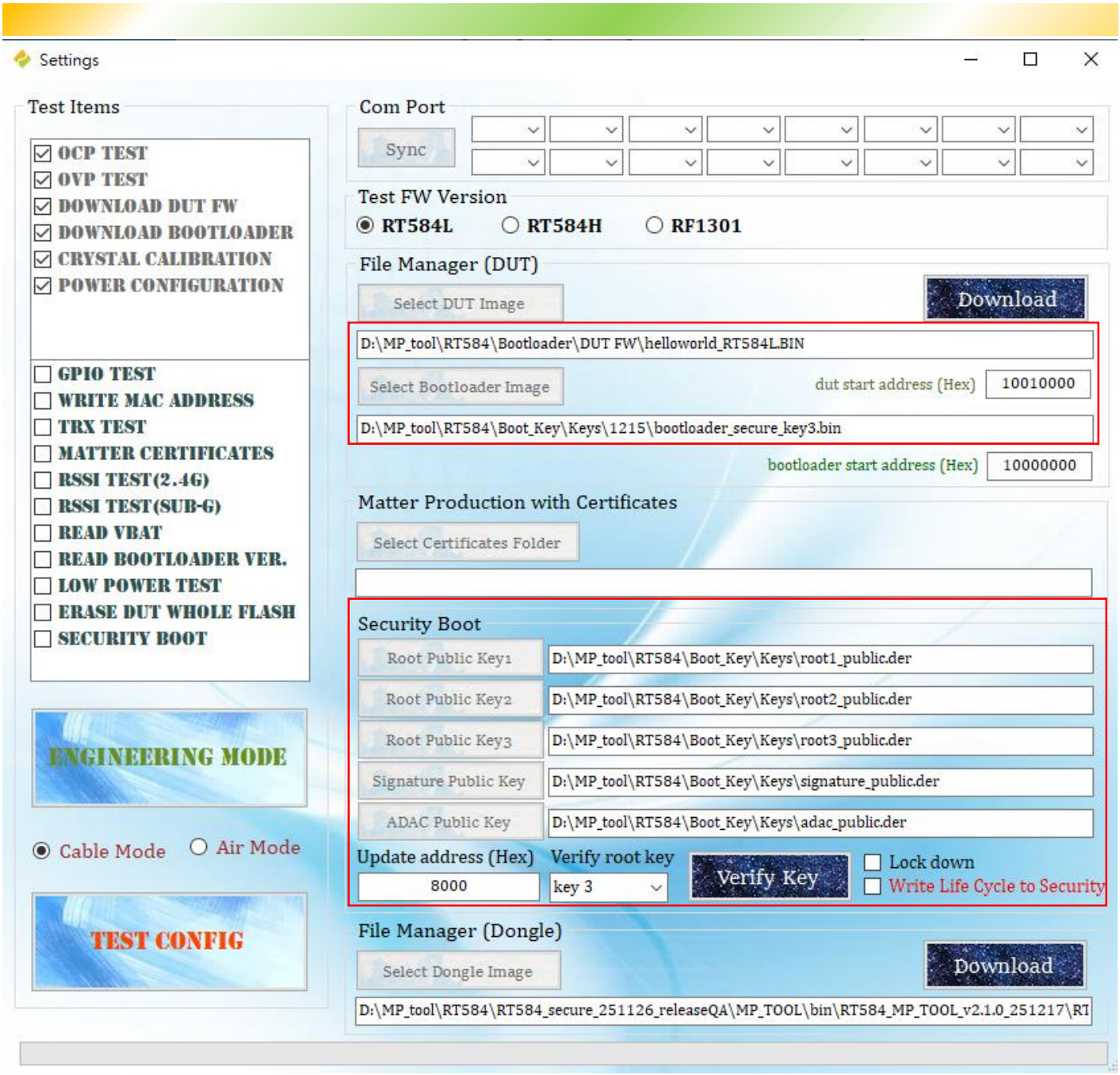


Figure 72. Verify bootloader with public key.

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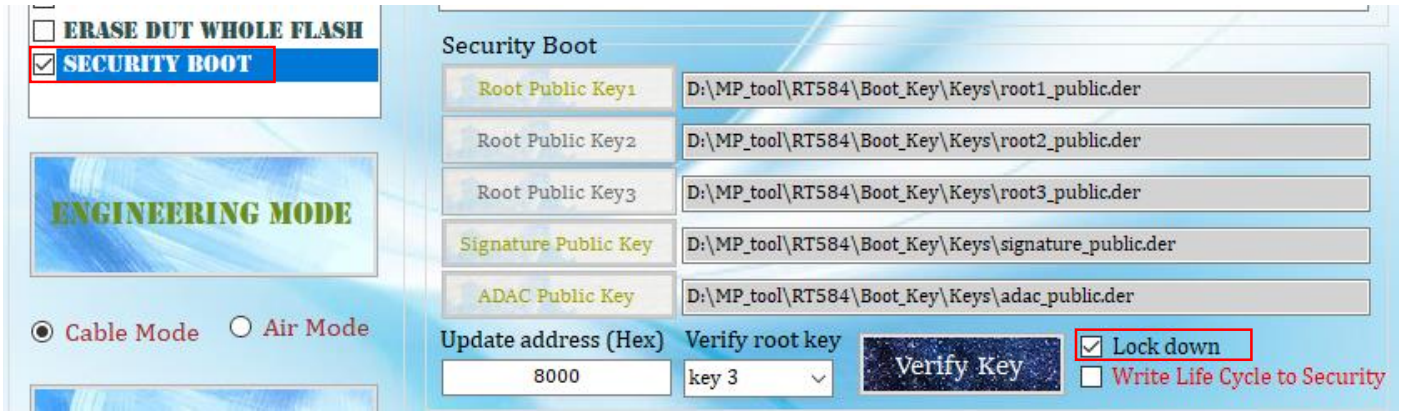


Figure 73.check the test item - SECURITY BOOT

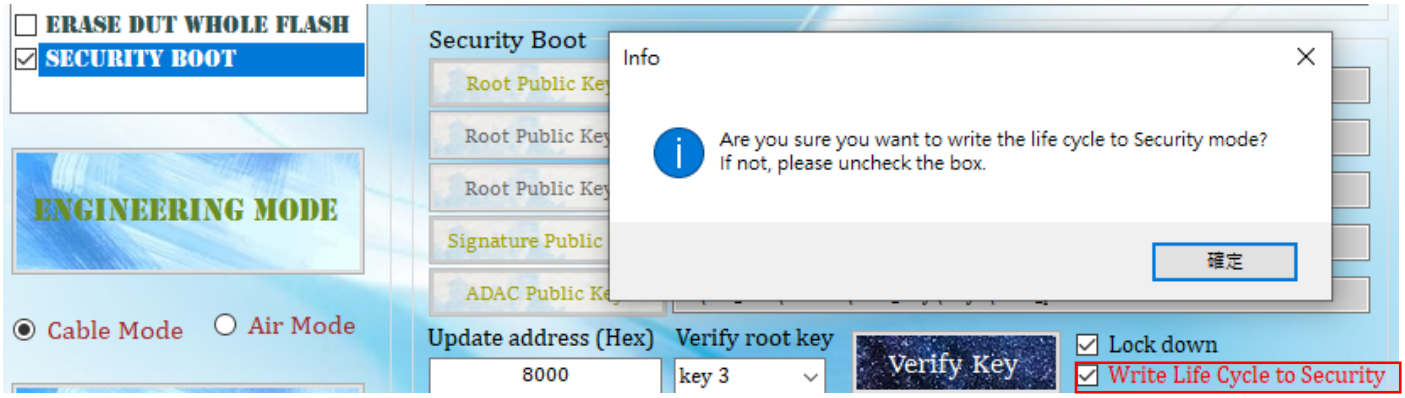


Figure 74. Check "Write Life Cycle to Security"

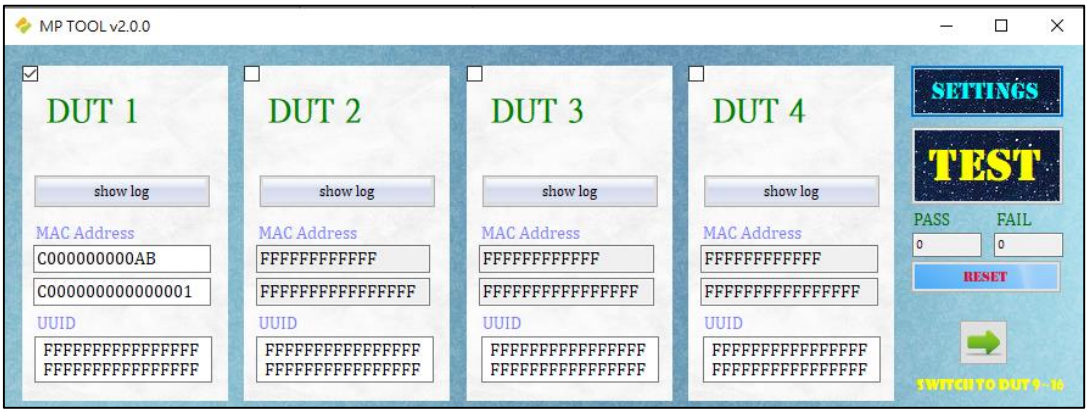


Figure 75. Click the TEST button.

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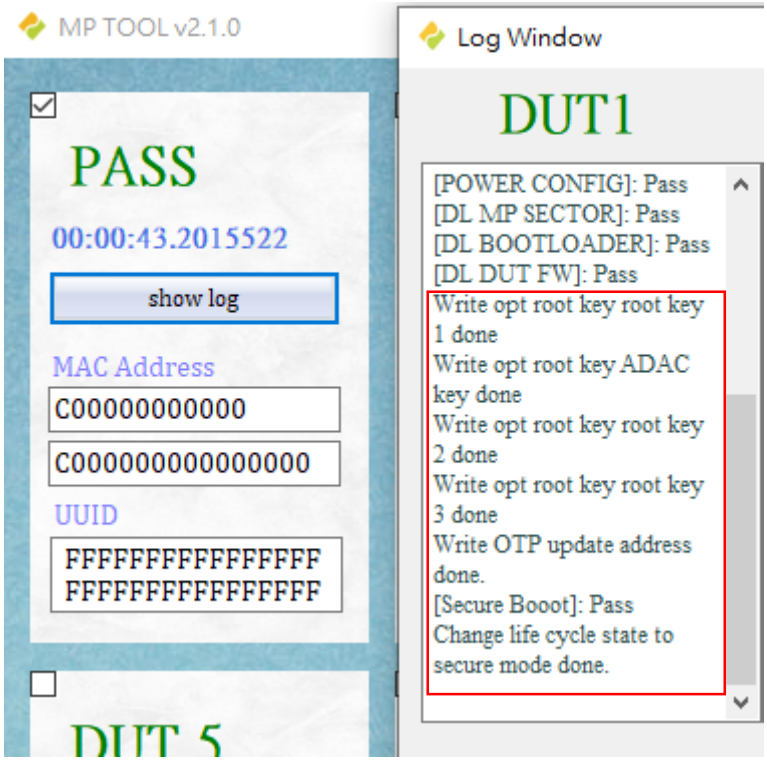


Figure 76. Show log of the ERASE DUT FLASH

Revision History

Revision	Description	Owner	Date
2.0	Initial version.	Kathy Lin	2025/08/01
2.1	Add functions - Erase DUT whole flash, Read bootloader version and security boot.	Kathy Lin	2025/11/21

Note: For complete command details and usage examples, refer to the respective command sections in this document.

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